Stem Cells and Regenerative Medicine

An in vitro model for vitamin A transport across the human blood-brain barrier

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Abstract

Vitamin A, supplied by the diet, is critical for brain health, but little is known about its delivery across the blood-brain barrier (BBB). Brain microvascular endothelial-like cells (BMECs) differentiated from human-derived induced pluripotent stem cells (iPSC) form a tight barrier that recapitulates many of the properties of the human BBB. We paired iPSC-derived BMECs with recombinant vitamin A serum transport proteins, retinol binding protein (RBP) and transthyretin (TTR), to create an *in vitro* model for the study of vitamin A (retinol) delivery across the human BBB. iPSC-derived BMECs display a strong barrier phenotype, express key vitamin A metabolism markers and can be used for quantitative modeling of retinol accumulation and permeation. Manipulation of retinol, RBP and TTR concentrations, and the use of mutant RBP and TTR, yielded novel insights into the patterns of retinol accumulation in, and permeation across, the BBB. The results described herein provide a platform for deeper exploration of the regulatory mechanisms of retinol trafficking to the human brain.

eLife assessment

This **fundamental** work substantially advances our understanding of retinol transport through the blood-brain barrier. The evidence supporting the conclusions is **compelling**, with rigorous biochemical assays. In general, the work is of broad interest to cell biologists, biochemists and neuroscientists.

Introduction

Retinoids (Vitamin A and related compounds) are essential micronutrients supplied by the diet that regulate over 500 genes (1) and are involved in vision, embryonic development, cell differentiation, metabolism and brain health (2–5). Retinoids play critical roles in both brain development and maintenance of brain health. For example, retinoic acid is a key signaling compound that induces neural differentiation and acquisition of unique brain vasculature properties in the prenatal brain (6,7). In the mature brain, retinoids contribute to maintenance of synaptic plasticity and sleep regulation; vitamin A deficiencies can lead to learning and memory deficits and depression of long-term potentiation (8–11). Vitamin A levels (circulating principally as retinol) naturally diminish with time, and there is mounting evidence that patients with Alzheimer's disease (AD) have lower serum levels of vitamin A than age-matched controls, (12,13) and that retinoid deficiencies contribute to cognitive decline in AD (12,14,15). Retinoids regulate expression of several genes that have been implicated in AD (14), specifically those genes involved in generation (16–22) and clearance (23–25) of the AD-related peptide beta-amyloid. Reciprocally, AD pathology may disrupt normal vitamin A trafficking (25,26).

The mobilization of retinol (ROH) stored in the liver is well characterized. Briefly, ROH is packaged with retinol-binding protein 4 (RBP) (27) in hepatocytes and then secreted into the blood. There, the ROH-RBP complex binds a second protein, transthyretin (TTR), thereby preventing clearance of the complex in the kidney (28). The distribution of ROH across its primary protein transporters in the blood is shown in **Figure 1A**.

In order to exert biological activity, ROH must cross from the blood into target tissues. There remain several unsettled questions regarding the mechanisms of vitamin A uptake (see (29)). ROH is lipophilic, and several *in vitro* studies demonstrate that 'free' ROH (not bound to RBP) can enter lipid vesicles and cross bilayers through diffusion (30–33). Due to its lipophilicity, ROH binds serum albumins non-specifically, with K_D's reported between 200 to 7600 nM (34,35); however, in humans serum albumins are not believed to transport ROH *in vivo* (36), nor participate directly in transport of ROH across the cell membrane. Other studies provide evidence that cell-surface protein(s) participate in RBP-mediated ROH transport across cell membranes (37–40), of which one has been identified as the transmembrane protein STRA6 (Signaling Receptor and Transporter of Retinol) (41). STRA6 binds RBP, mediates ATP-independent bi-directional transfer of ROH (42–45) and is involved in cell-signaling (46–48). It is well established that ROH must be released from RBP to enter the cell via STRA6 and that RBP is not internalized (37,38). ROH uptake is enhanced through coupling with cellular retinol binding protein 1 (CRBP1), the primary intracellular ROH binding protein in most tissues (44,49), and lecithin retinol acyltransferase (LRAT), the enzyme chiefly responsible for esterification of ROH into retinyl esters (RE), the "long-term storage" form of vitamin A (41,42). Recent cryo-EM characterization of STRA6 provides structural evidence that STRA6 may mediate ROH transfer from both free lipid and/or ROH-RBP complexes (50), but no functional data were provided. STRA6 can mediate transfer of ROH from preparations of ROH-RBP-TTR (41,49,51); however, ROH-RBP-TTR does not appear to directly bind cell surfaces (51), which suggests dissociation of ROH-RBP complex from TTR is required prior to ROH delivery.

The blood brain barrier (BBB) is the major site of nutrient exchange between the brain and circulation (52). The primary barrier phenotype is imparted by brain microvascular endothelial cells (BMECs), which are heavily polarized: the luminal (blood-facing) and abluminal (brain-facing) membranes differ significantly in lipid and protein composition (53). The mechanism(s) by which vitamin A is accumulated at the BMEC luminal membrane and then permeated across the BMEC abluminal membrane remains largely unexplored, in large part due to the inherent difficulty in conducting controlled *in vivo* studies. A diagram of suspected ROH delivery modes across the BBB is shown in **Figure 1B**.

Investigation into the mechanism of ROH transport to the brain would be greatly facilitated by well-characterized *in vitro* models of the human BBB. Recently, techniques have been developed for reprogramming human-derived cells into an induced pluripotent state (iPSCs) which can be differentiated into cell types that are difficult to obtain from primary tissues (54,55). Of relevance to this work, iPSCs have been differentiated into brain microvascular endothelial-like cells (BMECs) (56,57). These iPSC-derived BMEC-like cells express many markers of *in vivo* BMECs, and they are robust and readily produced. Critically, iPSC-derived BMEC-like cells show transcriptional expression of STRA6 (57), suggesting vitamin A uptake at the luminal BBB

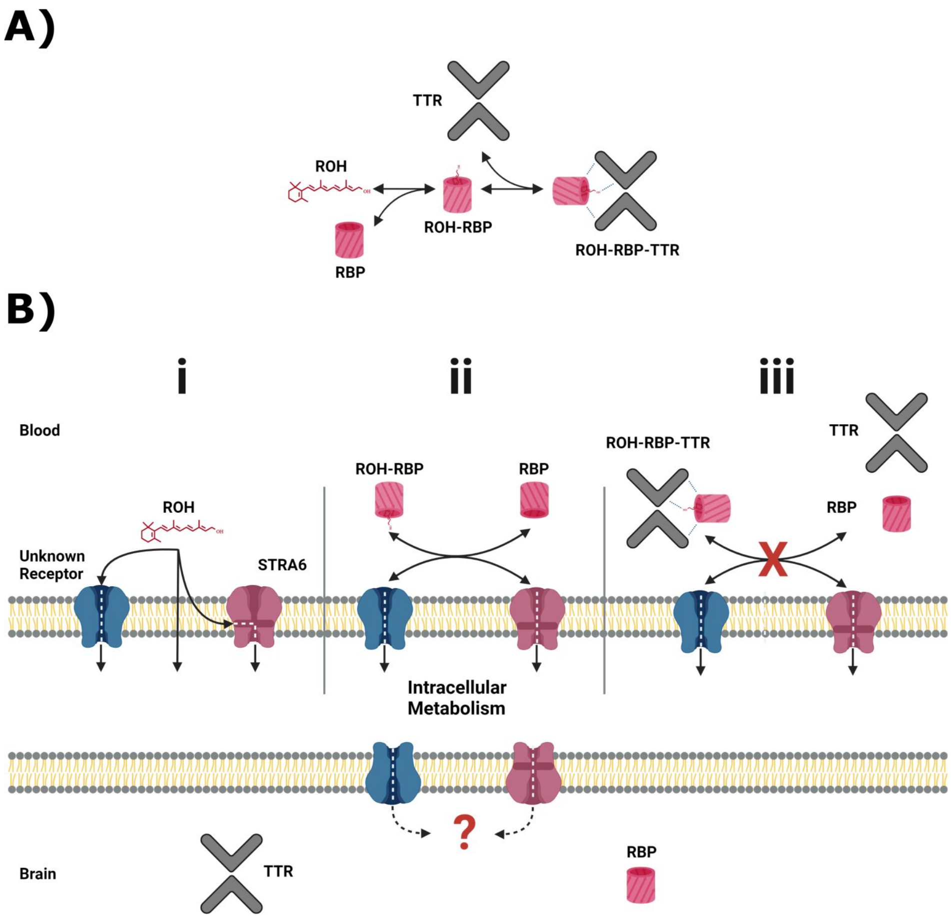

Figure 1

Primary serum vitamin A carriers and potential routes of delivery at the BBB.

A) Principal retinol distribution between free-lipid and protein-bound states in the blood. Retinol (ROH) in the blood partitions between free-lipid and protein-complexed states, with retinol binding protein (RBP) and transthyretin (TTR) serving as the principal serum ROH transporters. **B)** Putative ROH delivery mechanisms at the blood brain barrier (BBB). Critically, RBP and TTR do not cross the BBB. **(i)** In the blood, free ROH may cross the BBB by lipophilic diffusion through the lipid bilayer or through a specific cell-surface transporter, such as STRA6. **(ii)** ROH in complex with RBP (ROH-RBP) is known to deliver ROH to cell-surface transporters, including STRA6. STRA6 is thought to mediate exchange of retinoids between blood and intracellular pools. The net accumulation or release of ROH is thereby dependent on the ratio of ROH-RBP and unbound RBP present in the blood. **(iii)** ROH-RBP is complexed with TTR as it circulates in the blood. The ROH-RBP-TTR complex has not been shown to directly bind to cell-surface transporters, including STRA6, suggesting that dissociation of ROH-RBP from TTR is required before ROH is internalized as shown in panel (ii). Regardless of the entry method, all ROH is thought to enter the intracellular retinoid metabolism. It is unknown how intracellular retinoids are transported into the brain. RBP and TTR are expressed in the brain and presumably transport ROH throughout the brain.

membrane may be facilitated by STRA6 in a similar manner to other cell types. However, the mechanism of vitamin A permeation at the BBB abluminal membrane remains wholly unexplored.

In this report, we demonstrate the feasibility of using human iPSC-derived BMEC monolayers cultured on plasticware or permeable Transwells in conjunction with recombinant human RBP and TTR as an *in vitro* platform for studying retinoid uptake by, and transport across, the BBB. We have recently reported a robust method for producing and purifying recombinant human RBP in both apo- and holo-forms (58) as well as recombinant human wild-type (59) and mutant TTR (60). To support the validity of this *in vitro* BBB model for retinoid transport studies, we confirmed expression of STRA6, LRAT and CRBP1 protein in our iPSC model. We then investigated ROH accumulation and permeation across iPSC-derived BMECs when delivered by the two primary physiologic sources: RBP and RBP-TTR complex. To further explore the utility of this experimental model, we compared accumulation and permeation of free lipid ROH to that of RBP-bound ROH. We also mutated RPB (L63R/L64S) and TTR (I84A) to modify their binding properties, providing a means to examine the role of each protein individually in ROH accumulation and permeation. Finally, we probed BMEC response to ROH uptake as a function of delivery mechanism by quantifying RNA expression levels for STRA6, LRAT and CRBP1. Our results establish the utility of this platform for obtaining greater mechanistic insight into retinol trafficking across the blood-brain barrier.

Results

iPSC-derived BMECs express BBB relevant phenotypes and key retinoid-related proteins

We verified that induced pluripotent stem cell-derived (iPSC) brain microvascular endothelial-like cells (BMECs) display BBB-relevant markers using commercially available antibodies (**Table M1**), consistent with prior work (56,57). Specifically, iPSC-derived BMECs express **(Figure 2A)**: endothelial cell marker PECAM-1; tight junction markers CLDN5, OCLN and ZO-1; GLUT1, a glucose transporter highly enriched at the BBB *in vivo*. Additionally, these cells express efflux transporters, including BCRP and MRP1, as expected (**Figure 2B**).

We next tested whether BMECs express key transporters and enzymes involved in retinoid metabolism and / or transport using commercially available polyclonal antibodies (**Table M1**). Critically for our purpose, differentiated BMECs express STRA6 (**Figure 2B**), corroborating transcriptional evidence (57). Notably, STRA6 expression does not appear uniform. BMECs also express CRBP1, with a staining pattern consistent with CRBP1's cytosolic function (61); similarly, BMECs express LRAT, with faint staining in the lipid bilayer and stronger staining near the cell nucleus consistent with LRAT's function in retinyl ester synthesis in lipid droplets (62,63). Immunocytochemistry results for STRA6, CRBP1 and LRAT were corroborated by Western blotting using the same polyclonal antibodies as for the immunocytochemistry panel (**Table M2**, **Figure 2C**), with detection of bands at ~16 kDa (CRBP1), ~30 kDa (LRAT) and ~70 kDa (STRA6) as expected. In order to assuage concerns about commercially available antibodies against STRA6 (64), purified recombinant GST-tagged STRA6 protein was used as a positive control.

Recombinant RBP and TTR are suitable replacements for serum sources

We previously expressed, purified and characterized human wild-type transthyretin (TTR) (59) and human wild-type retinol binding protein 4 (RBP) (58) in *E. coli*. Recombinant RBP binds retinol (ROH) with a dissociation constant K_D = 100 ± 30 nM, indistinguishable from serum-derived

Total ROH Concentration (μM)	Delivery Mode[2]	ROH Distribution Based on K_D (μM)[1] Free ROH	ROH-RBP	ROH-RBP-TTR	Notes
0.1	Free	0.1	0	0	
0.4	Free	0.4	0	0	
2	Free	2	0	0	
2	ROH-RBP	0.4	1.6	0	ROH partitions between free and RBP-bound states.
2	ROH-muRBP	0.4	1.6	0	muRBP binding affinity to TTR and possibly to STRA6 is abolished.
2	ROH-RBP-TTR	0.14	0.18	1.68	ROH partitions between free, RBP-bound and RBP-TTR-bound states.
2	ROH-RBP-muTTR	0.4	1.6	0	ROH partitions between free and RBP-bound states. RBP does not bind to muTTR.

[1]Distribution of ROH between unbound (free) and protein-bound states was calculated by assuming equilibrium and utilizing the known or measured K_D for binding of RBP to ROH and for binding of TTR to ROH-RBP.

[2]Delivery mode indicates whether ROH was supplied to the cell culture medium in the absence of binding protein (free) or pre-complexed with RBP, muRBP (L63R/L64S), RBP-TTR or RBP-muTTR (I84A). Total RBP and TTR concentrations were 2 μM and 4 μM, respectively.

Table 1

ROH and protein concentrations for BMEC accumulation and permeability assays

Antibody Target	Manufacturer	Catalog Number	Clonality	Host	Dilution	Fixative
PECAM1	ThermoFisher	RB-10333-P1	Polyclonal	Rabbit	1 : 25	MeOH
CLDN5	ThermoFisher	35-2500	Monoclonal	Mouse	1 : 200	MeOH
OCLN	ThermoFisher	33-1500	Monoclonal	Mouse	1 : 50	MeOH
TJP1 (ZO-1)	ThermoFisher	33-9100	Monoclonal	Mouse	1 : 200	MeOH
SLC2A1 (GLUT1)	ThermoFisher	MA1-37783	Monoclonal	Mouse	1 : 500	MeOH
ABCG2 (BCRP)	Millipore Sigma	MAB4155	Monoclonal	Mouse	1 : 50	4% PFA
ABCC1 (MRP1)	Millipore Sigma	MAB4100	Monoclonal	Mouse	1 : 25	MeOH
STRA6	Abnova	H00064220-D01P	Polyclonal	Rabbit	1 : 200	4 % PFA
CRBP1	ThermoFisher	PA5-28713	Polyclonal	Rabbit	1 : 500	MeOH
LRAT	ThermoFisher	PA5-38556	Polyclonal	Rabbit	1 : 250	MeOH

Table M1

Antibodies and staining information for immunocytochemistry

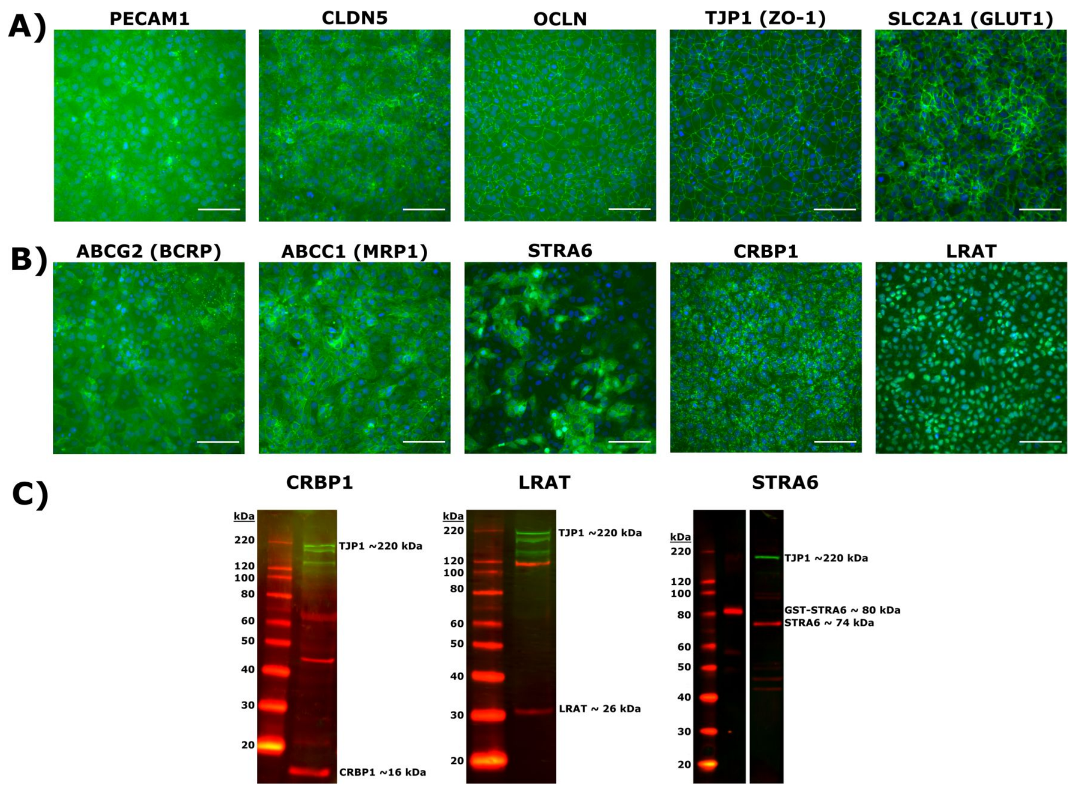

Figure 2

iPSC-derived BMEC marker and retinoid-related protein validation by immunocytochemistry and Western blot.

A) Expression of endothelial cell marker PECAM-1; tight junction and associated proteins CLDN5, OCLN and TJP1 (ZO-1); glucose transporter SLC2A1 (GLUT1). Proteins are labeled in green and nuclear stain in blue. Scale bar equals 100 µm. **B)** Expression of efflux transporters ABCG2 (BCRP), ABCC1 (MRP1) and retinoid-related proteins STRA6, CRBP1 and LRAT. Proteins are labeled in green and nuclear stain in blue. Scale bars equal 100 µm. **C)** Western blots of CRBP1 (red), LRAT (red), and STRA6 (red) confirming antibody specificity. An antibody against TJP1 (green) is used as a BMEC specific loading control in each blot. Polyclonal STRA6 antibody was additionally validated against recombinant GST-tagged STRA6.

Antibody Target	Manufacturer	Catalog Number	Clonality	Host	Dilution
LRAT	ThermoFisher	PA5-38556	Polyclonal	Rabbit	1 : 500
CRBP1	ThermoFisher	PA5-28713	Polyclonal	Rabbit	1 : 1000
STRA6	Abnova	H00064220-D01P	Polyclonal	Rabbit	1 : 1000
TJP1 (ZO-1)	ThermoFisher	33-9100	Monoclonal	Mouse	1 : 1000
TJP1 (ZO-1)	ThermoFisher	40-2200	Polyclonal	Rabbit	1 : 1000

Table M2

Antibodies and staining information for Western blot

human RBP (58). Recombinant RBP complexed with ROH (ROH-RBP) binds recombinant human TTR with a dissociation constant of ~ 250 nM, in close agreement with human serum-derived measurements (58).

ROH accumulates in iPSC-derived BMEC monolayers from ROH-RBP or ROH-RBP-TTR complexes

BMEC monolayers cultured on 96-well plates were exposed to solutions of ROH-RBP or ROH-RBP-TTR prepared at biologically relevant concentrations: 2 µM ROH (typical range *in vivo*, 1 – 2 µM (12,13)), 2 µM RBP (*in vivo* range, 2 – 4 µM (62)), and 4 µM TTR (*in vivo* range, 3 – 8 µM (65)). Solutions of ROH-RBP were prepared by overnight equilibration of free lipid ROH (1:20 ratio of ^{3}H-ROH to unlabeled ROH) and ligand-free (apo) RBP. Solutions of ROH-RBP-TTR were prepared by overnight equilibration of ROH-RBP (holo) and TTR. After the desired incubation time for accumulation, cells were washed, lysed, and the tritium signal counted. ROH cellular accumulation, in µmoles of ROH per L cell volume, was calculated from DPM measurements using the manufacturer supplied specific activity of ^{3}H-ROH, the ratio of ^{3}H-ROH to unlabeled ROH (1:20) and a cell volume of 1.37 x 10^{-12} L / cell. Cell volume was estimated by multiplying the average area of a BMEC (402 µm^{2}) by its height (3.4 µm). BMEC area was estimated in two ways: directly from ICC image analysis and by counting the number of cells in a BMEC monolayer after singularization and dividing by the culture dish area. The two methods produced consistent estimates of cell area. BMEC height was estimated by analysis of Z-stack ICC images. We use the term “ROH cellular accumulation” to represent the quantity of all added ROH that becomes cell-associated, recognizing that we did not determine whether any retinol was converted to oxidized metabolites or retinyl esters and we did not differentiate cell-associated radioactivity between internalized versus membrane-associated material after washing.

ROH cellular accumulation increased throughout the two-hour experiment at a steady rate with either ROH-RBP or ROH-RBP-TTR (**Figure 3A** and **Figures S1D, S1F**). TTR did not affect ROH cellular accumulation kinetics. After two hours, accumulated ROH cellular concentration was ~100 µM, or about 50-fold higher than the 2 µM ROH medium concentration. This indicates that the BMEC monolayer stores excess ROH and that ROH accumulates against a concentration gradient. High cellular ROH accumulation could be accounted for by a number of established mechanisms, including thermodynamically driven partitioning of the lipophilic ROH into lipid-rich cellular components, binding of ROH by intracellular proteins such as CRPB1, and/or storage of internalized ROH as retinyl esters (RE).

BMEC monolayers are a useful in vitro BBB model system for measuring ROH permeation at physiologically relevant conditions

iPSC-derived BMEC monolayers have been shown to be a good *in vitro* model for the permeability of essential nutrients and drugs across the BBB (56,57). Vitamin A permeability has not previously been measured with iPSC-derived BMEC monolayers, and in fact there is a very sparse literature on retinol transport across the BBB in any *in vitro* or *in vivo* model (66–68). BMECs were cultured on semi-permeable Transwell inserts that allow for sampling of the apical chamber (‘blood’) and basolateral chamber (‘brain’) (**Figure 3B**). Transendothelial electrical resistance (TEER) was used to confirm the integrity of the BMEC barrier. TEER measures the resistance of the monolayer to an electrical current and correlates with the size of molecules excluded from paracellular transport. For compounds the size of sucrose or retinol (342 Da or 286 Da, respectively), a minimum TEER of ~ 500 Ω x cm^{2} is considered sufficient to exclude paracellular transport (69). TEER for iPSC-derived BMECs in our study was typically 3000 Ω x cm^{2} at the start of incubations, and remained above 1000 Ω x cm^{2} upon conclusion.

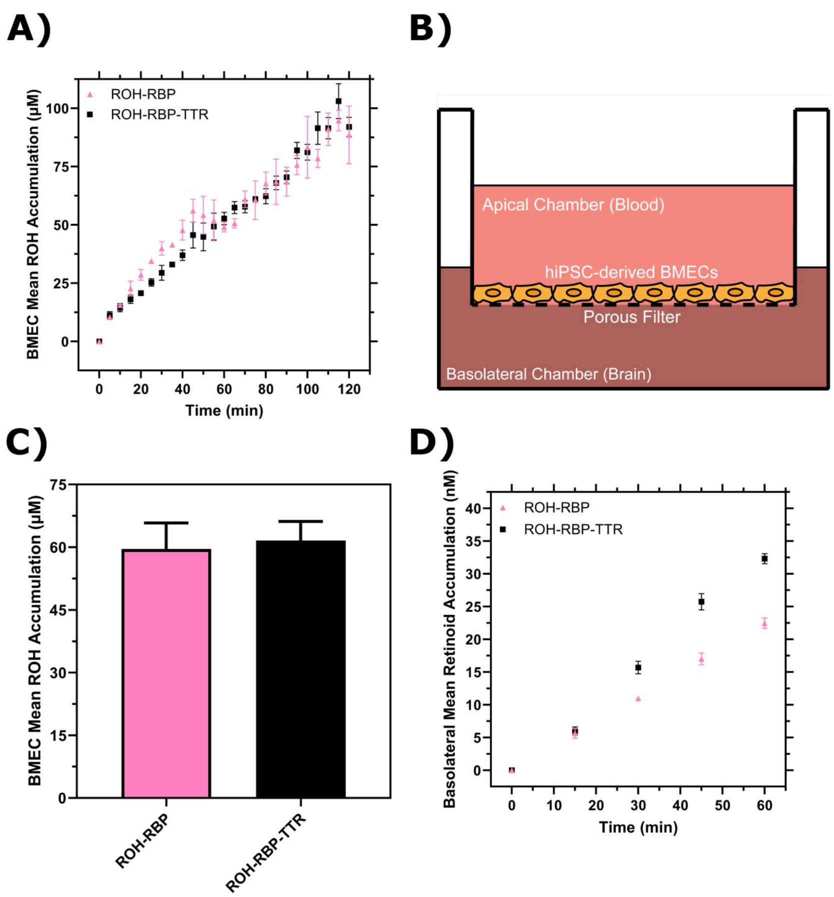

Figure 3

BMEC ROH uptake and permeation mediated by RBP or RBP-TTR complex.

A) Mean ROH cellular accumulation as a function of time. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the average cell volume. Error bars represent the standard deviation of three biological replicates. Fluid concentrations are typical of human blood concentrations at 2 µM ROH, 2 µM RBP and 4 µM TTR. **B)** Schematic of the Transwell apparatus. The semi-permeable support allows for BMEC basolateral efflux. **C)** Mean ROH cellular accumulation in BMEC lysate after 60 minutes, collected from cells in Transwells. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the calculated cell volume. **D)** Kinetics of retinoid accumulation in the basolateral chamber. Error bars represent the standard deviation of four biological replicates. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the volume of the basolateral chamber medium. Apical concentrations were 2 µM ROH, 2 µM RBP and 4 µM TTR. No RBP or TTR was added to the basolateral chamber.

ROH-RBP or ROH-RBP-TTR solutions (using ^{3}H-ROH as a tracer) were charged to the apical chamber at concentrations identical to the accumulation assay, and samples from both apical and basolateral chambers were collected at several time points over the course of the one hour experiment. We describe accumulated ^{3}H signal in the basolateral chamber as retinoid, recognizing that we were not able to collect enough material to confirm its chemical identity specifically as retinol. ^{14}C-sucrose was added to all samples as a control for barrier integrity since sucrose does not cross the *in vivo* BBB in large quantities (57). In these experiments there is no RBP or TTR added to the basolateral chamber. At the conclusion of each experiment, BMEC monolayer lysate was collected to test for closure of the mass balance; ~97% of ^{3}H-ROH and at least 95% of ^{14}C-sucrose radioactivity was recovered (**Table S1 and S2**). ^{3}H-ROH signal measured in lysates from Transwell experiments was used to calculate total cellular accumulation at the one hour time point (**Figure 3C**); cellular accumulation of ~60 μM is consistent with the monolayer results at one hour (**Figure 3A**) and again shows no statistical difference between ROH-RBP and ROH-RBP-TTR.

As shown in **Figure 3D**, there was a short lag period after which retinoid accumulation in the basolateral chamber increased linearly over the course of the one-hour experiment. Interestingly, although ROH cellular accumulation was not affected by TTR in the apical chamber (**Figures 3A and 3C**), the basolateral retinoid accumulation was roughly 30% higher when TTR was present in the apical chamber. Basolateral retinoid accumulation after 1 hour was only a small fraction (1-1.5%) of the ROH concentration loaded in the apical chamber.

To quantify our data and to confirm basolateral accumulation was due to permeability across the cellular monolayer rather than via paracellular leakage, we used **Eq S1** and **Eq S2** to calculate the apparent permeability (Pe_{app}) of the BMEC monolayer and Transwell semi-permeable insert combined for both sucrose and ROH (**Table S3**). Sucrose Pe_{app} ranged from 0.58 ± 0.03 to 0.69 ± 0.04 x 10^{-6} cm/s when mixed with ROH-RBP-TTR or ROH-RBP samples, respectively, in agreement with prior studies using these iPSC-derived BMECs (Pe = 0.57 x 10^{-6} cm/s) (57) and slightly lower (indicating a tighter barrier) than reported with primary porcine BMECs (Pe = 1 x 10^{-6} cm/s, TEER < 1000 Ω x cm^2) (67). These results confirm tight barrier integrity in this iPSC-derived model system. Pe_{app} for ROH was 4.8 ± 0.2 x 10^{-6} cm/s when supplied by ROH-RBP and 6.6 ± 0.3 x 10^{-6} cm/s when supplied by ROH-RBP-TTR. These permeabilities are an order of magnitude higher than those for sucrose and in the same range as that of glucose (Pe = 3.7 x 10^{-6} cm/s (57)), a critical nutrient for the brain. This analysis indicates that ROH is indeed transported transcellularly, and that inclusion of TTR increases the ROH permeation rate by ~30%. To our knowledge, this is the first report of permeability measurements for RBP-bound ROH across a BBB-like monolayer, as well as the first reported indicator that TTR may play a role in increasing permeation of ROH across the BBB.

Free ROH cellular accumulation is bulk fluid concentration-dependent

In the presence of RBP and/or TTR, ROH partitions between free and protein-bound states. Although ROH circulating with RBP is thought to be the predominant mode of vitamin A delivery to cells and tissues, there is evidence that free ROH readily partitions into cell membranes (30–33). The estimated concentrations of free and protein-bound ROH at our experimental conditions were calculated from the known equilibrium dissociation constants and total concentrations of ROH, RBP and TTR (**Table 1**).

For RBP, about 20% of ROH is free (0.4 μM free versus 1.6 μM bound to RBP), while when both RBP and TTR are present, only about 7% (0.14 μM) of ROH is estimated to be free. The relative contribution of free ROH to overall cellular accumulation and permeation across barriers is not known and has typically not been accounted for in other studies of RBP-mediated ROH delivery to cells (41 ,43 ,44 ,49 ,51 ,64 ,70 ,71), although there is evidence cells respond differently to free ROH as compared to ROH-RBP (29).

We used our experimental system to measure the free ROH concentration-dependent cellular accumulation and to compare the data to cellular accumulation data from protein-bound ROH. BMEC monolayers were exposed to ROH at three concentrations: 2 μM (physiological), 0.4 μM (to approximate the free ROH concentration in ROH-RBP solutions), and 0.1 μM (to approximate the free ROH concentration in ROH-RBP-TTR solutions). Cell-associated ROH increased over the two-hour time course of the experiment, with total accumulation a strong function of fluid-phase ROH concentration (**Figure 4A and Figure S1A-C**). The accumulated cellular ROH concentration after 2 h was nearly two orders of magnitude higher than the fluid-phase ROH concentration, consistent with the data observed for protein-bound ROH (**Figure 3A**). To examine concentration-dependent patterns of ROH accumulation kinetics, cellular concentrations were normalized by the ROH concentration initially loaded in the medium (**Figure 4B**), as the bulk fluid concentrations remained relatively stable over the course of the experiment (**Table S4**). This analysis demonstrates that the kinetic pattern is distinctly different at each ROH fluid-phase concentration. Briefly, at 0.1 μM ROH in the fluid phase, cellular concentration reached ~9 μM (or a cell:fluid ratio of ~ 90 μM/μM) by approximately 90 minutes, beyond which it remained stable. At 0.4 μM fluid-phase ROH, cellular concentration reached an apparent plateau of ~36 μM (cell:fluid ratio of ~ 90 μM/μM) at approximately 60 minutes, but then started to increase further at approximately 90 minutes. In contrast, at 2 μM fluid-phase ROH, cellular concentration increased continuously over the two-hour experiment, reaching ~320 μM (~160 μM/μM cell/fluid concentration ratio).

We evaluated whether these data could be described by a simple kinetic model, where we assume the bulk concentration remains constant (see **Table S5**). The strong concentration dependence suggested a partitioning model (akin to solvent:solvent partitioning) as a better descriptor than a receptor-ligand binding model. If c_f = bulk fluid concentration (μM), c_{cell} = accumulated cellular concentration (μM) at any time t, k_1 = first-order rate constant (min^{-1}), K_p = partition coefficient (μM cell/ μM fluid), then a simple model is:

$$\frac{c_{cell}}{c_f} = K_p[1 - exp(-k_1 t)] \qquad \text{Eq. 1}$$

However, given the apparent intermediate plateau for the 0.4 μM free ROH sample, as well as the observation that the ratio c_{cell}/c_f increases with increasing fluid-phase concentration, we hypothesized that a secondary uptake mechanism is triggered after a lag time, t_{lag}, that corresponds to crossing an intracellular ROH threshold, $c_{cell}{}^*$:

$$\frac{c_{cell}}{c_f} = K_p[1 - exp(-k_1 t)] + K_p{}^* \left[1 - exp\left(-k_1{}^*\left(t - t_{lag}\right)\right)\right] \qquad \text{Eq. 2}$$

where

$$t_{lag} = {-ln\left[1 - \frac{c_{cell}{}^*}{K_p c_f}\right]}\Big/{k_1}$$

We fit the data by **Eq. 1** (**Figure S2**) or **Eq. 2** (**Figure 4B**) using least-squares regression. Our analysis indicates that including the secondary uptake mechanism provides a significantly better description of the experimental data (**Figure S2** and **Table S5**), providing support for the hypothesis of a biphasic response at higher fluid ROH concentrations. A possible explanation is that loading of CRBP1 with ROH to its capacity (estimated from our modeling to be $c_{cell}{}^*$ ~ 36 μM) triggers initiation of a secondary storage mechanism, such as retinyl ester synthesis, to handle

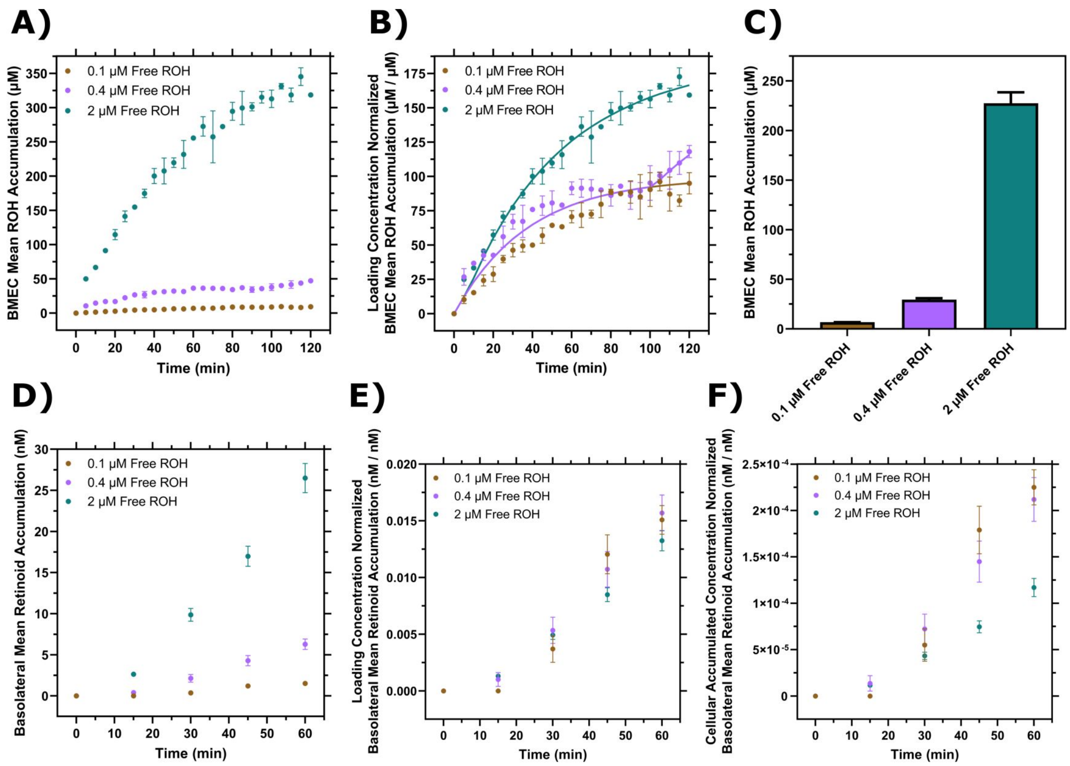

Figure 4

BMEC ROH uptake and permeation as a function of free ROH concentration.

A) Mean ROH cellular accumulation as a function of time and free ROH concentration. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the average cell volume. Error bars represent the standard deviation of three biological replicates. Fluid concentrations were 0.1, 0.4, or 2 µM ROH. No RBP or TTR was present in the medium. **B)** Mean ROH accumulation from Panel A normalized by the ROH concentration in the medium. Data are fit by a simple partitioning model with a secondary uptake mechanism (**Eq. 2**) that triggers upon accumulation exceeding a fitted threshold value of ~36 µM. **C)** Mean ROH cellular accumulation in BMEC lysate after 60 minutes, collected from cells in Transwells. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the calculated cell volume. No RBP or TTR was added to either chamber. **D)** Kinetics of retinoid accumulation in the basolateral chamber. Error bars represent the standard deviation of four biological replicates. Measured DPM values were converted to accumulated concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the volume of the basolateral chamber medium. No RBP or TTR is added to either chamber. **E)** Retinoid basolateral chamber accumulation normalized by the apical chamber ROH concentration. **F)** Retinoid basolateral chamber accumulation normalized by the accumulated cellular ROH concentration at 60 minutes.

additional ROH cellular accumulation. Indeed, biphasic ROH uptake from RBP in Sertoli cells has been reported; the accumulated ROH at the plateau in that study corresponded to the intracellular CRBP1 concentration (72).

Free ROH permeates across the BBB-like barrier with kinetics proportional to apical concentration and not cellular concentration

Permeation of protein-free ROH at 0.1 μM, 0.4 μM or 2 μM in the apical chamber was measured using the Transwell setup. To ensure barrier integrity, TEER and ^{14}C-sucrose permeability were monitored as described previously. At the conclusion of each experiment, BMEC monolayer lysate was collected to test for closure of the mass balance; at least 90% of ^{3}H-ROH and at least 96% of ^{14}C-sucrose radioactivity was recovered (**Table S1 and S2**). We first confirmed that cellular accumulation in this setup was consistent with that in the monolayer by measuring radioactivity in the lysate at the end of the permeation experiment (**Figure 4C**). As shown in **Figure 4D**, basolateral retinoid accumulation rate increased with increasing apical ROH concentration. We calculated Pe_{app} of ROH to range from 3.8 ± 0.4 to 7.7 ± 0.5 x 10^{-6} cm/s as the apical ROH concentration increased from 0.1 to 2 μM. These values indicate that ROH is about 10-fold more permeable than sucrose (0.43 ±0.03 to 0.58±0.03 x 10^{-6} cm/s) and are in good agreement with data reported for free ROH (supplied at ~10^{-8} M) in primary porcine BMECs (Pe = 4.1 ± 0.7 x 10^{-6} cm/s) (67). The permeabilities are similar to those we calculated for ROH-RBP or ROH-RBP-TTR, demonstrating that RBP is not required for ROH transit across the BBB-like barrier. It is important to note that Pe_{app} should be independent of the apical ROH concentration for a single permeation mechanism. The observation that Pe_{app} increases with increasing fluid phase ROH concentration supports the hypothesis that higher fluid phase ROH concentrations trigger secondary transport mechanisms in BMECs or that free ROH is processed intracellularly into different retinoid forms, such as retinyl esters.

To understand the concentration dependence of ROH permeation, we normalized the basolateral ROH concentration by the initial apical ROH concentration or by the accumulated cellular ROH concentration (**Figures 4E and 4F**). The data collapse to a single curve in **Figure 4E** but not in **Figure 4F**, demonstrating that transport of retinoid across the BMEC monolayer to the basolateral chamber is more tightly coupled to the apical chamber ROH concentration rather than to the accumulated concentration of ROH within the cell barrier.

RBP and TTR markedly reduce ROH cellular accumulation compared to protein-free ROH, but permeability across the BBB-like barrier is similar for protein-bound and protein-free ROH

We asked whether cellular ROH accumulation kinetics differed between free ROH and protein-bound ROH when supplied at the same initial total ROH concentration. Accumulation kinetics for 2 μM free ROH (**Figure 4A**) are compared to accumulation kinetics for 2 μM ROH-RBP (**Figure 3A**) in **Figure 5A**. Since the calculated free ROH concentration in the 2 μM ROH-RBP sample at these conditions is ~ 0.4 μM (**Table 1**), data at 0.4 μM ROH (**Figure 4A**) are also included for comparison. ROH cellular accumulation levels from 2 μM ROH-RBP are substantially lower than at 2 μM free ROH, but slightly exceed that of 0.4 μM free ROH alone. For samples containing TTR (**Figure 5B**), the free ROH concentration is estimated to be ~0.14 μM (**Table 1**). ROH cellular accumulation levels from 2 μM ROH-RBP-TTR (**Figure 3A**) are markedly lower than those at 2 μM free ROH, but significantly exceed that at 0.1 μM free ROH (**Figure 4A**).

Cellular uptake of ROH in protein-bound samples could be achieved through two paths operating in parallel: directly from free ROH and/or by delivery from ROH bound to RBP or RBP-TTR. We wondered whether we could estimate the relative importance of these two paths by comparing

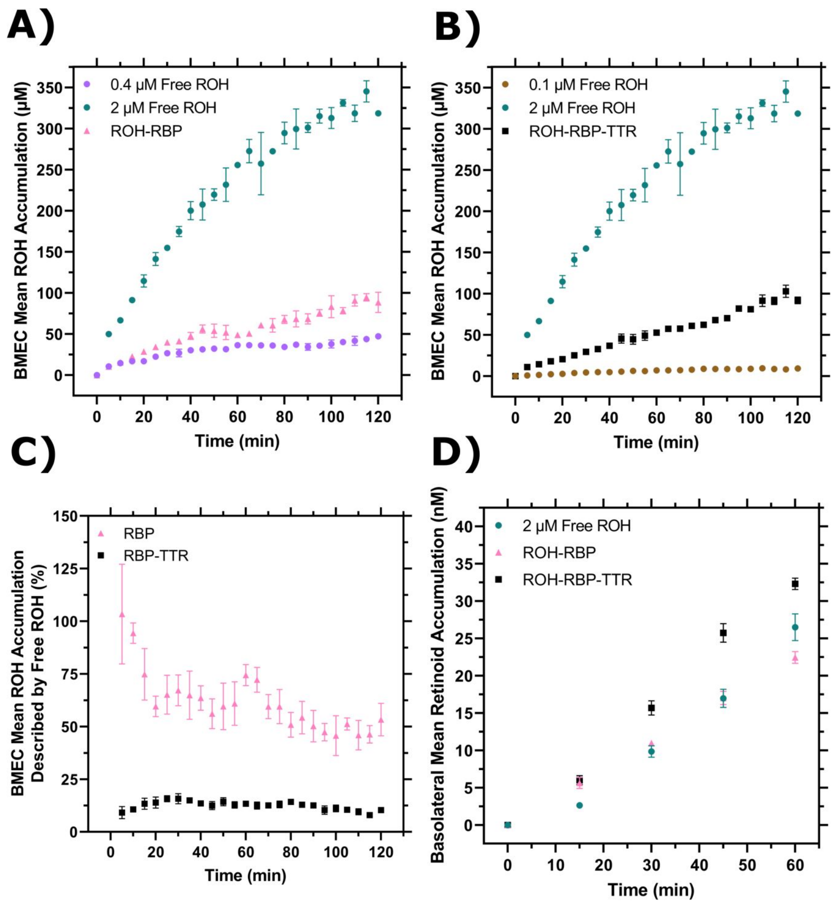

Figure 5

Comparison of cellular ROH accumulation and permeation with or without RBP or TTR.

A) Cellular accumulation from ROH-RBP, replotted from **Figure 3A**, compared to 0.4 µM or 2 µM ROH, replotted from **Figure 4A**. The ROH-RBP solution is an equilibrated mixture of 2 µM ROH and 2 µM RBP, with a calculated protein-free (unbound) ROH concentration of 0.4 µM. **B)** Cellular accumulation from ROH-RBP-TTR, replotted from **Figure 3A**, compared to 0.1 µM or 2 µM ROH, replotted from **Figure 4A**. The ROH-RBP-TTR solution is an equilibrated mixture of 2 µM ROH, 2 µM RBP, and 4 µM TTR, with a calculated protein-free (unbound) ROH concentration of 0.14 µM. **C)** Percentage of cellular ROH accumulation from ROH-RBP or ROH-RBP-TTR that could be potentially described by the contribution of free ROH present in the protein samples (0.4 µM free ROH for ROH-RBP and 0.1 µM for ROH-RBP-TTR). **D)** Comparison of basolateral retinoid permeation for free ROH, ROH-RBP and ROH-RBP-TTR samples each containing 2 µM total ROH per sample.

cellular accumulation data from ROH alone versus ROH complexed to RBP or RBP-TTR (**Figure 5C**). Specifically, at each time point we normalized the free ROH cellular accumulation by the total ROH cellular accumulation measured for the protein-bound samples (0.4 µM free ROH normalized by ROH-RBP for RBP alone samples and 0.1 µM free ROH normalized by ROH-RBP-TTR for TTR-containing samples – see **Table 1**). For ROH-RBP, essentially 100% of the initial observed ROH cellular uptake could be accounted for by the contribution of the 0.4 µM free ROH present in the mixture. This percentage drops to 50% after 2 hours, as ROH cellular accumulation from ROH-RBP samples grows more quickly than for 0.4 µm free ROH alone. This analysis is consistent with the hypothesis that free ROH partitions rapidly into the cell, but then that pathway's contribution diminishes over time as tightly controlled influx / efflux via the ROH-RBP route becomes more dominant. In contrast, ROH cellular accumulation from the 0.1 µM free ROH in ROH-RBP-TTR samples never exceeds ~20% of the total uptake. Thus, although the kinetics of cellular accumulation are virtually identical between ROH-RBP and ROH-RPB-TTR (**Figure 3A**), the *relative* contributions of free versus protein-delivered ROH appear to be quite different (**Figure 5C**). This illustrates the importance of careful accounting for all bound and free forms of ROH in mechanistic investigations.

Although *cellular* accumulation was much higher for 2 µM free ROH than when bound to RBP or RBP-TTR, permeation into the *basolateral* chamber was similar (but not identical) for all three cases, again highlighting that permeation across the BMEC barrier is correlated more strongly with the total apical ('blood') ROH concentration and not with the free apical ROH concentration or the cellular ROH accumulated concentration (**Figure 5D**). However, basolateral permeation was not entirely a function of total apical ROH concentration, but also depended on whether ROH was presented with RBP alone or in complex with TTR; permeation with RBP alone was lower than for an equivalent total concentration of free ROH, whereas permeation with RBP in complex with TTR was higher than for an equivalent concentration of free ROH. This observation suggests that RBP and TTR might influence ROH cellular trafficking even though the proteins are not themselves internalized, and that the proteins may have subtle differences in the observed effect.

RBP and TTR mutants reveal novel insights into mechanisms of ROH cellular accumulation in, and permeation across, BMEC monolayers

To further explore the role RBP and TTR play individually in ROH cellular accumulation and permeation, we generated mutants that abrogate wild-type binding interactions. L63R/L64S mutations in RBP (muRBP) alter a loop region at the entrance of the β-barrel of RBP (**Figure 6A**, adapted from (73)), and this loop region is critical in mediating the binding interaction to TTR. Furthermore muRBP has been previously shown unable to bind to a protein expressed on placental membranes that participates in retinol transport (74); it is suspected, but has not been confirmed, that this protein is STRA6. We produced and purified muRBP, and purity was confirmed using analytical size exclusion chromatography (aSEC) and DEAE anion exchange chromatography (**Figure S3**). RBP and muRBP elute in one primary peak on both aSEC and DEAE columns. On aSEC, muRBP elutes earlier than RBP, suggesting the L63R/L64S mutation causes an increase in the apparent volume of the protein. However, refolded ROH-muRBP displays an $A_{330/280}$ ratio of ~ 1.0 in the presence of excess ROH, indicating normal 1:1 binding stoichiometry (data not shown). Furthermore, ROH binds to muRBP with K_D = 110 ± 60 nM, comparable to RBP (**Figure 6B**). Interestingly, muRBP fluorescence at saturation is only ~ 50% the magnitude of ROH-RBP, which we theorize is due to changes in the β-barrel local environment that alter the efficiency of resonance energy transfer; this hypothesis would be consistent with the apparent larger volume seen by aSEC. To confirm loss of TTR-binding affinity, we utilized a fluorescence anisotropy assay that demonstrates ROH-muRBP does not bind TTR (**Figure 6D**).

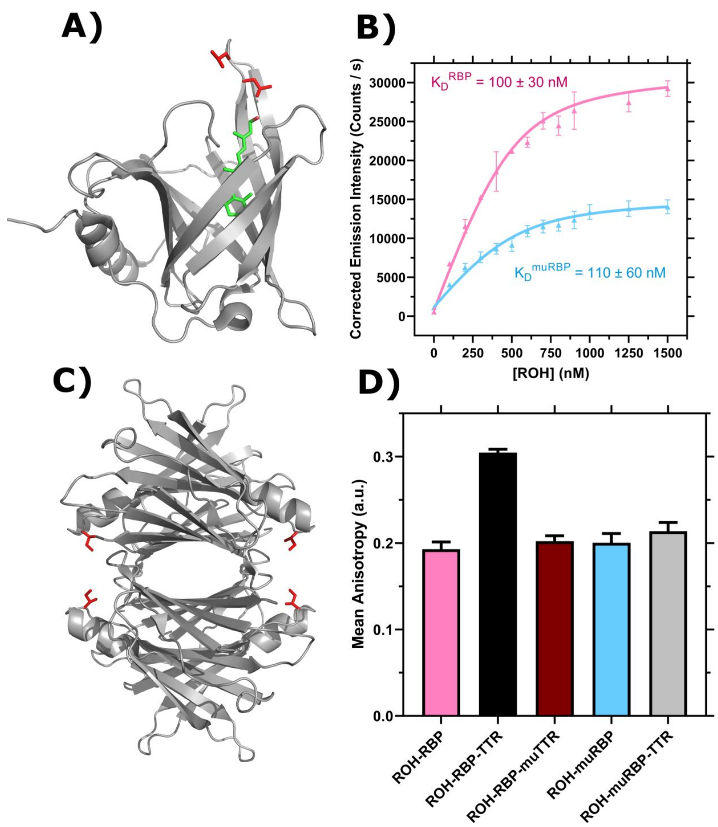

Figure 6

Characterization of RBP, TTR and mutants.

A) PDB Entry 5NU7 (73). Ribbon diagram displays RBP with bound ROH shown in green and L63 and L64 in red. **B)** ROH binding to RBP (pink, adapted from (58)) and muRBP (blue) monitored by emission of ROH at 460 nm as an acceptor in resonance energy transfer from donor RBP tryptophan. Data are fit by non-linear regression as described (58). Error bars represent the standard deviation of three independent replicates. **C)** PDB Entry 5CN3 (75). Ribbon diagram displays the TTR tetramer. I84 for each TTR monomer is shown in red. **D)** Representative binding of 1 μM ROH-RBP or 1 μM ROH-muRBP to 4μM TTR or 4μM muTTR as measured by fluorescence anisotropy using the polarized emission of ROH at 460 nm. Error bars represent the standard deviation of three technical replicates.

To examine the role of TTR binding to RBP on retinol uptake and trafficking, we produced an I84A mutant TTR (muTTR) with reduced affinity for RBP (28,76) (**Figure 6C**, adapted from (75)). Fluorescence anisotropy confirmed that recombinant muTTR did not bind measurably to ROH-RBP (**Figure 6D**).

First, we measured ROH cellular accumulation in BMEC monolayers using muRBP or muTTR and compared the results against RBP or TTR. We found negligible differences in ROH cellular accumulation between ROH-RBP and ROH-RBP-muTTR (**Figure 7A**); this result was expected since muTTR does not bind ROH-RBP.

Surprisingly, accumulation from ROH-muRBP is markedly *higher* than that of ROH-RBP; indeed, accumulation approaches that of free ROH at 2 μM. This was an unexpected result given that muRBP binds ROH with similar affinity as RBP, but may be consistent with the hypothesis that muRBP cannot participate in ROH efflux from intracellular ROH stores, and thereby drives a higher net ROH accumulation.

We next measured ROH permeability across the BMEC monolayer in the Transwell configuration using muRBP or muTTR with RBP. We confirmed consistency of the monolayer plate and Transwell experiments, showing again that cellular ROH accumulation was significantly higher with muRBP than with RBP (**Figure 7B**). Although *cellular* accumulation was much higher for ROH bound to muRBP than compared to RBP, *basolateral* retinoid permeation was virtually identical (**Figure 7C**). This result is consistent with our other observations that basolateral permeation is primarily correlated with apical and not cellular ROH concentration.

Because the I84A TTR mutant does not bind to RBP, we anticipated that ROH-RBP-muTTR would demonstrate permeability similar to ROH-RBP alone. However, basolateral permeability was higher in samples containing TTR or muTTR when compared to ROH-RBP alone (**Figure 7C**). This surprising result indicates that it is the presence of TTR, and not specifically the binding of TTR to RBP, that is responsible for the higher basolateral permeation of ROH.

TTR and muTTR upregulate expression of LRAT

It is known that influx of ROH delivered by ROH-RBP through STRA6 triggers an intracellular signaling cascade (46,48). STRA6-mediated transport of ROH is coupled to intracellular CRBP1 (47,49) concentrations, while LRAT plays an important role in storing excess intracellular ROH by enzymatically converting it to retinyl esters. We used our *in vitro* system to explore whether ROH uptake may trigger transcriptional changes in these vitamin A-related proteins when treated with ROH and its binding partners. Specifically, we used RT-qPCR to measure changes in expression of STRA6, CRBP1 and LRAT after treatment of BMECs with the ROH preparations listed in **Table 1**. Primer sequences used are detailed in **Table M3**.

Neither STRA6 nor CRBP1 expression was affected to a statistically significant degree by addition of ROH, RBP or TTR (**Figure 8A** and **Figure 8B**). Nor was there a significant change in LRAT expression when BMECs were exposed to free ROH, ROH-RBP, or ROH-muRBP (**Figure 8C**). Samples containing either ROH-RBP-TTR or ROH-RBP-muTTR, however, show statistically significant upregulation of LRAT (**Figure 8C**). Since TTR binds to RBP but muTTR does not, LRAT upregulation is independent of TTR binding to RBP and is decoupled from the delivery of ROH through STRA6 via RBP. Since LRAT plays a role in managing intracellular ROH inventory, this result could explain why the presence of TTR, and not its binding to RBP, increases ROH permeability across the BMEC monolayer (**Figure 7C**). Although more detailed investigation is needed, our data are indicative of a heretofore unknown function for TTR in regulating ROH delivery across barriers, independent of its well-known role as a carrier for RBP.

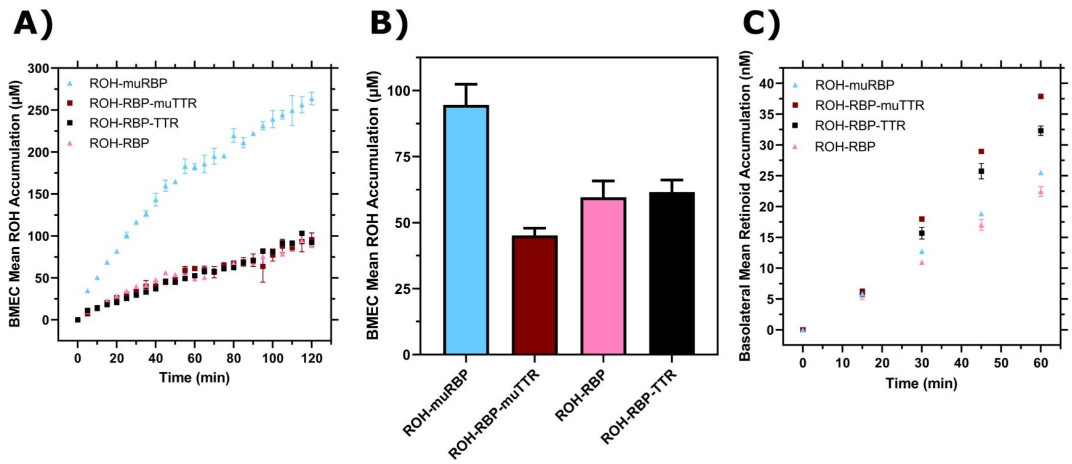

Figure 7

Comparison of RBP and TTR to muRBP and muTTR.

A) Mean ROH cellular accumulation as a function of time and concentration. Error bars represent the standard deviation of three biological replicates. The medium contains 2 μM ROH equilibrated with 2 μM RBP, 2 μM muRBP, 2 μM RBP + 4 μM TTR or 2 μM RBP + 4 μM muTTR. **B)** Mean ROH cellular accumulation in BMEC lysate after 60 minutes, collected from cells in Transwells. Apical concentrations are the same as listed in Panel A. **C)** Kinetics of retinoid permeation into the basolateral chamber comparing ROH-RBP, ROH-muRBP, ROH-RBP-TTR and ROH-RBP-muTTR.

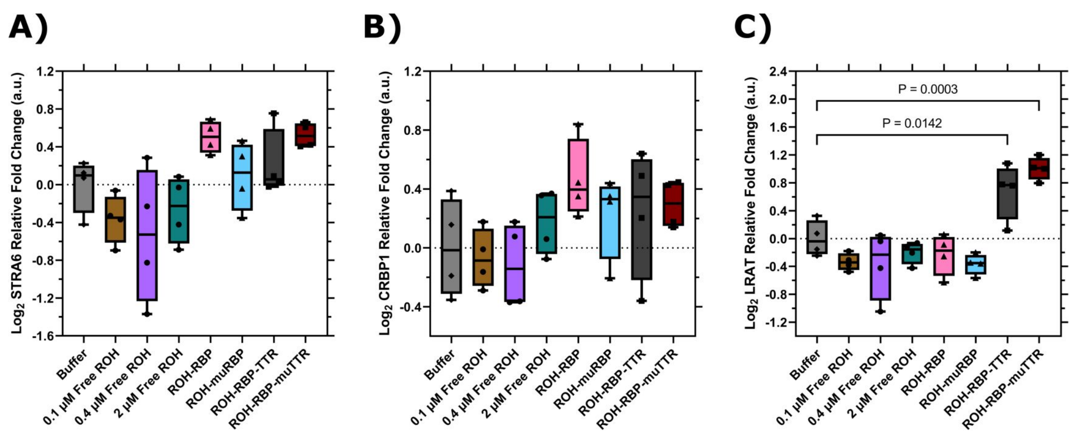

Figure 8

RT-qPCR data for BMECs treated with different ROH modalities for two hours.

Expression values are normalized to the housekeeping gene, ACTB, and quantified relative to BMECs treated with HBSS alone. ΔΔCq data are presented in box-and-whisker format after $\log_2$ transformation with the values for each biological replicate displayed individually (*N = 4*). Statistical analyses were performed in Prism on $\log_2$ transformed ΔΔCq values via one-way ANOVA followed by Dunnett's test using a confidence interval of 95%. **A)** STRA6; **B)** CRBP1; **C)** LRAT.

Discussion

Despite the importance of retinoids to brain health, very little is known about the mechanisms by which retinoids enter the specialized cells forming the blood-brain barrier, or how retinoids transit the BBB. Using human iPSC-derived BMECs, as well as recombinant human RBP and TTR, we constructed an *in vitro* platform that provides a practical means for controlled experimentation and quantitative interrogation of retinol uptake by, and delivery across, the BBB. iPSC-derived BMECs express markers indicative of brain endothelial specificity, including PECAM1, Claudin-5, Occludin, ZO-1 and Glut1. BCRP and MRP1 expression provide evidence of efflux transporter activity, a major function of the *in vivo* BBB. In this work we show that BMECs express STRA6 as well as two other vitamin-A relevant proteins: LRAT and CRBP1, both of which are known to couple with STRA6 for ROH uptake. Recombinant RBP and TTR were produced that are suitable replacements for human plasma-derived materials while providing a means to probe the roles of these proteins individually in ROH trafficking and transport via specific mutations. Using this *in vitro* system, we measure for the first time the rate of ROH transport across a human BBB-like barrier and we show that ROH permeability is similar to that of an essential brain nutrient, glucose.

Our results clearly show that cells take up free ROH in a concentration-dependent manner, and that ROH binding proteins are not required for cellular accumulation (**Figure 4A**). After two hours, cell-associated ROH concentrations exceed that in the fluid phase by two orders of magnitude. Since transport of free ROH into the cell is maintained against a strong concentration gradient, this result indicates that iPSC-derived BMECS store ROH primarily in a bound or associated form. This is similar to another blood-barrier, the retinal pigment epithelium (RPE), where 92% of total intracellular ROH is stored as retinyl esters (RE) and 8% as ROH-CRBP1 complex (77).

The kinetic pattern of BMEC ROH uptake varied with ROH concentration (**Figure 4B**). The biphasic behavior of ROH uptake in Sertoli cells is similar to what we observed in iPSC-BMECs at intermediate concentrations of ROH; the plateau in that study occurred at an intracellular ROH concentration corresponding to the CRBP1 concentration (72). We propose that, at 0.1 μM fluid phase ROH, BMECs maintain an adequate sink for intracellular ROH, possibly through binding to CRBP1. At higher ROH concentrations, ROH accumulates intracellularly beyond the CRBP1 binding capacity, and a secondary storage system is recruited into action. A simple kinetic model was developed that is consistent with this proposed mechanism and provides an estimate of the 'triggering' cellular ROH concentration as ~36 μM. The secondary storage system may proceed through LRAT, which normally esterifies ROH bound to CRBP1 and is regulated by the ratio of bound to unbound CRBP1; however, in periods of excess free ROH, LRAT can act on free ROH directly (78,79). Further study is needed to validate this hypothesis.

Not only can free vitamin A be taken up by BMECs, but it also can be transported across the barrier without requiring RBP or TTR (**Figure 4D**), see **Figure 1B**. Notably, basolateral ('brain') accumulation was directly proportional to the apical ('blood') ROH concentration, and *not* to the cellular concentration. This suggests that cells at the BBB store excess ROH and carefully regulate retinoid delivery to the 'brain' in response to blood concentrations, and/or that protein-delivered ROH has greater ability to access trans-cellular trafficking pathways than ROH taken up as free lipid.

RBP carries out two key functions: it sequesters ROH in the blood in order to maintain free ROH concentrations at sub-toxic levels, and it delivers ROH to cells via STRA6. We tested whether ROH accumulation in, or retinoid transport across, BMECs was enhanced by RBP. Notably, complexation of ROH with RBP *slows* cellular accumulation considerably relative to what occurs in the absence of RBP at an equivalent ROH concentration (compare ROH-RBP versus 2 μM free ROH, **Figure 5A**). Interpretation of this result requires consideration of the fact that there is ~0.4 μM free

ROH in the ROH-RBP preparation, and a better comparison therefore should account for the contribution of the 0.4 μM free ROH present in the ROH-RBP preparation. We hypothesize that cellular uptake of ROH occurs by two parallel paths, from free ROH and from ROH-RBP. A direct comparison of the kinetics (**Figure 5C**) leads to our hypothesis that free ROH rapidly partitions into the cell, whereas the RBP-mediated cellular accumulation is slower; after two hours, however, about half of the total ROH cellular concentration may be accounted for by each of the pathways.

Experiments with the L63R/L64S mutant RBP provide further insight into the role of RBP in regulating ROH trafficking. Cellular uptake of ROH is significantly faster with muRBP compared to RBP, and accumulation reached levels approaching that of 2 μM free ROH. This was an unexpected result, since muRBP binds ROH with equal affinity to RBP (**Figure 6B**). As illustrated in **Figure 1B**, ROH can cross cellular membranes through multiple mechanisms: by passage of free ROH through the lipid bilayer, by passage through cell-surface proteins, or by STRA6-mediated release of ROH from ROH-RBP. Importantly, RBP-mediated transport through STRA6 is bidirectional, with both influx and efflux depending on availability of intracellular apo-CRBP1 and extracellular apo-RBP, respectively (43). Furthermore, ROH efflux requires binding of apo-RBP to STRA6 (43,49), a mechanism available to RBP but presumably not muRBP (74). We postulate that cellular accumulation is therefore the net sum of three steps: uptake of free fluid-phase ROH, uptake of ROH via delivery from holo-RBP, and efflux of cellular ROH via uptake by extracellular apo-RBP. The substantially higher cellular accumulation for muRBP compared to RBP provides support for the hypothesis that *efflux* of ROH is a critical regulatory component for BMEC intracellular ROH levels and that this pathway is inoperable with muRBP. Additional support for this hypothesis would require direct characterization of the binding of RBP and muRBP to our iPSC-derived BMECs, a subject for future research. Regardless, these data suggest that the most important role of RBP during times of retinol abundance is to mediate efflux of excess intracellular ROH, thereby maintaining intracellular ROH at sufficient, but not excessive, concentrations.

Despite the large differences in ROH cellular accumulation with 2 μM free ROH versus the same quantity incubated with RBP or muRBP, the ROH permeation kinetics were very similar (**Figure 7C**). This result suggests that BMEC monolayers regulate transport of ROH across the barrier by 'sensing' the apical ('blood') ROH concentration. A plausible mechanism for achieving 'sensing' is via the intracellular ratio of ROH-bound CRBP1 and ligand-free apo-CRBP1. Apo-CRBP1 promotes retinyl ester (RE) hydrolysis (79,80) and influx of ROH through STRA6 (43), while ROH-CRBP1 promotes RE synthesis via LRAT (78) and efflux of ROH through STRA6 (43). Efflux of ROH through STRA6 requires availability of ligand-free RBP (apo-RBP) in the blood. Therefore, the ratio of intracellular ROH-CRBP1 to apo-CRBP1 is directly coupled via STRA6 to blood ROH through the ratio of extracellular ROH-RBP to apo-RBP.

TTR's primary role in ROH transport is generally believed to be in binding to RBP and preventing loss of the small protein through the kidney (28). ROH is more buried in the ROH-RBP-TTR complex compared to ROH-RBP (28), and cross-linking studies show evidence of an ROH-RBP-STRA6 complex but not of an ROH-RBP-TTR-STRA6 complex (51); taken together, these observations suggested that dissociation of ROH-RBP from TTR is required for RBP-mediated transfer of ROH to STRA6. Indeed, we observed that ROH cellular accumulation with ROH-RBP-TTR was indistinguishable from ROH-RBP (**Figure 3A**), which at first glance would indicate that TTR does not play a direct role in ROH uptake into the BMEC monolayer (see **Figure 1B, iii**). However, there are significant differences in the ROH distribution between ROH-RBP and ROH-RBP-TTR delivery modalities that may suggest a more complicated scenario: first, the free ROH concentration is ~ 4x lower when both RBP and TTR are present (~0.14 μM, see **Table 1**) than compared to RBP alone (0.4 μM, see **Table 1**); second, the concentration of ROH-RBP differs by an order of magnitude between RBP and TTR containing samples (0.18 μM, see **Table 1**) and the RBP alone samples (1.6 μM, see **Table 1**). The comparative analysis shown in **Figure 5C** indicates that direct uptake of free ROH in ROH-RBP-TTR mixtures contributes much less to the overall cellular accumulation than when compared to ROH-RBP mixtures alone. Since there is no evidence of direct binding of ROH-RBP-TTR to STRA6, we postulate that ROH-RBP in ROH-RBP-TTR mixtures must play a much larger role in ROH cellular influx than when ROH-RBP is presented alone. This may have significant physiologic ramifications; STRA6-mediated JAK/STAT signaling is activated specifically by ROH *influx* via ROH-RBP (see **Figure 1B, ii**) (47); furthermore, this signaling cascade requires ROH to be delivered directly from RBP, as neither RBP nor ROH alone induce the cascade (46).

When TTR was added to the apical chamber, we observed higher permeation into the basolateral chamber, and a higher Pe_{app} than that of free ROH or ROH-RBP at the same total apical ROH concentration (**Figure 7C**). Surprisingly, this effect was seen with both TTR and muTTR, despite the fact that muTTR does not bind to ROH-RBP. This result indicates that it is the presence of TTR, and not the binding of TTR to RBP, that is responsible for the higher permeation of ROH. To gain further insight, we looked for changes in expression of STRA6, CRBP1 or LRAT. No statistically significant changes in STRA6 or CRBP1 expression were detected after two hours of incubation. With LRAT, however, we saw a statistically significant increase in expression in BMECs exposed to either TTR or muTTR, but not to ROH or ROH-RBP alone. This is a novel finding that suggests TTR plays an important role in regulating retinol trafficking and transport across barriers. Moreover, this regulatory activity does not require TTR to be complexed to RBP. If TTR signaling directly increases expression of LRAT, concomitant enhanced production of RE could serve as the substrate for basolateral efflux, the mechanistic basis of which is wholly unknown. Further studies are required to tease out this unexpected role of TTR in retinol processing.

Taken together, our results demonstrate the utility of our *in vitro* BBB model constructed from iPSC-derived BMECs and recombinant wild-type and mutant RBP and TTR for studies of retinol trafficking across the BBB. Several novel findings include: (1) accumulation of ROH in the cells of the BBB is a strong function of the delivery mode (free or protein-bound), while permeation across the BBB is mostly independent of delivery mode, (2) retinol permeation rates across the BBB are similar to that of glucose, another essential brain nutrient, (3) efflux of ROH through STRA6 to apo-RBP in the serum may be an underappreciated route for controlling intracellular accumulation in times of retinol abundance, (4) TTR upregulates LRAT expression and influences ROH transport to the brain using mechanisms that are independent of its RBP-binding role. We highlight the importance of using wild-type and mutant RBP and TTR, as well as careful accounting for the distribution of ROH between free and protein-bound states, in any mechanistic investigation of retinol trafficking, permeability or STRA6-mediated signaling.

By leveraging the renewable, scalable and genetically-manipulable features of human iPSC-derived BMECs, the roles of key cellular proteins involved in retinoid accumulation, metabolism and permeation across the BBB could be directly explored. Recent advances in CRISPR-mediated perturbation methods, such as knockout ($CRISPR_{ko}$), interference ($CRISPR_i$) or activation ($CRISPR_a$), can be used to map the roles of STRA6, LRAT and/or CRBP1 in retinoid processing at the BBB. Results reported here lay the groundwork for more detailed investigations into mechanisms of retinol transport into the brain, which are expected to yield greater insights into retinol's role in supporting brain health and to provide novel approaches for treatment of retinol dysfunction in neurodegenerative disease.

Experimental Procedures

Induced pluripotent stem cell (iPSC) differentiation to brain microvascular endothelial-like cells (BMECs)

IMR90-4 iPSCs (WiCell, Madison, WI) were cultured on Matrigel-coated 6-well plates and supplemented daily with E8 medium (Stem Cell Technologies) as described (81). iPSCs were passaged in clumps at ~ 70% confluency every 3 – 5 days by dissociation with Versene (Life Technologies) at typical ratios between 1:6 and 1:12. To initiate differentiation into BMECs, iPSCs at ~ 70% confluency were dissociated and singularized by treatment with Accutase (Life Technologies) for 7 – 10 minutes and then diluted into fresh E8 media (1:4 *v/v*). Cells were counted on a hemocytometer, and then centrifuged at 200 x *g* for 5 minutes. The supernatant was aspirated and the pellet re-suspended in fresh E8 medium supplemented with 10 µM ROCK inhibitor (Tocris R&D Systems). Re-suspended cells were seeded at a density between 7,500 – 12,500 cells / cm^2 on fresh Matrigel-coated plates (Day – 3). ROCK-supplemented E8 media was aspirated and replaced with fresh E8 media (without ROCK inhibitor) 24 hours later to promote iPSC colony formation. Cells were subsequently expanded for 48 hours with daily E8 media replacement until reaching the optimal density of 30,000 cells / cm^2 as described previously (82). Unconditioned medium (UM: 50 mL knock-out serum replacement, 2.5 mL non-essential amino acids, 1.25 mL Glutamax (all from Life Technologies) and 1.75 µL of β-mercaptoethanol (Sigma) diluted into Dulbecco's Modified Eagle Medium / Nutrient Mixture F-12 (DMEM) to 250 mL and vacuum filtered into 0.2 µm PES filter-top bottles) was prepared and stored at 4 °C for up to two weeks. On Day 0, E8 medium was replaced with UM replenished daily for six days. On Day 6, UM media was replaced with fresh EC +/+ media for 48 hours without replacement. EC +/− medium was prepared as follows: 5 mL B-27 Supplement (50X), serum free (ThermoFisher) was diluted into human endothelial serum-free medium (Life Technologies) to 250 mL and vacuum filtered into 0.2 µm PES filter-top bottles. EC +/− medium was stored at 4 °C for up to two weeks. EC +/+ media was prepared fresh daily by supplementation of EC +/− media with 10 µM all-*trans* retinoic acid (Sigma, diluted in DMSO) and FGF2 (WiCell / Waisman Biomanufacturing) diluted 1:5000. On Day 7, plasticware (Corning) was coated with a 4:1:5 *v/v/v* collagen IV (Sigma) / fibronectin (Sigma) / sterile water stock solution. For Transwell-Clear permeable inserts (0.4 µm pore size), the concentration of collagen and fibronectin, respectively, was 400 µg/mL and 100 µg/mL. Each Transwell filter was coated with 200 µL of 4:1:5 solution. For all other plasticware, 4:1:5 solution was further diluted 5-fold in sterile water. On Day 8, cells were singularized with Accutase for 30 – 45 minutes, re-suspended in EC +/+ medium and plated onto the 4:1:5 collagen IV / fibronectin-coated plasticware prepared fresh the day before at a density of 1×10^6 cells / cm^2 for 1.12 cm^2 Transwell-Clear permeable inserts, at a density of 250,000 cells / cm^2 on 6/12-well tissue culture polystyrene plates, or at a density of 1×10^6 cells / cm^2 for 24/48/96-well tissue culture polystyrene plates. On Day 9, EC +/+ medium was replaced with EC +/− medium without retinoic acid to promote barrier tightening. At least one Transwell was seeded per differentiation to monitor transendothelial electrical resistance (TEER) as a measure of BMEC quality. TEER was measured every 24 hours after subculture on Day 8 to confirm barrier tightness. Resistance was recorded using an EVOM ohmmeter with STX2 electrodes (World Precision Instruments Sarasotae, FL). Maximum TEER values for iPSC-derived BMECs prepared by this protocol are typically > 2000 Ω × cm^2, calculated by multiplying the resistance readings by 1.12 cm^2 to account for the Transwell surface area. Maximum TEER was typically observed on Day 10, on which all subsequent experiments were performed.

Immunocytochemical analysis of BMECs

BMECs were singularized on Day 8 and seeded on 96-well plates coated with 4:1:5 solution. On Day 10, cells were rinsed once with Dulbecco's phosphate-buffered saline (DPBS) and fixed with either 100 % Methanol or 4 % paraformaldehyde for 10 – 15 minutes at room temperature. After fixation, the fixing agent was aspirated and the cells were washed three times in immediate succession with DPBS. After washing, cells were incubated for 60 minutes at room temperature in blocking buffer (10 % *v/v* goat serum in DPBS) before overnight incubation at 4 °C with primary antibodies diluted in blocking buffer (**Table M1**).

After primary incubation, cells were washed three times at 5 minutes each with DPBS before incubation for 1 hour at room temperature in the dark with secondary antibodies in blocking buffer (goat anti-rabbit IgG (H + L) Cross-Adsorbed Secondary Antibody conjugated to Alexa Fluor 488 (ThermoFisher, 1:200 dilution, Cat # A-11008) or goat anti-mouse IgG1 Cross-Adsorbed Secondary Antibody conjugated to Alexa Fluor 488 (ThermoFisher, 1:200 dilution, Cat # A-21121)). After secondary antibody incubation, cells were immediately stained for 15 minutes at room temperature in the dark with Hoechst nuclear count stain (ThermoFisher) diluted 1:5000 in DPBS. Cells were washed three times and visualized in fresh DPBS on a Nikon Ti2 epifluorescence microscope with a 20 X objective. Images were analyzed with FIJI software.

Western blot analysis for LRAT, CRBP1 and STRA6 expression

BMECs were singularized on Day 8 and seeded on 6-well plates coated with 4:1:5 solution. On Day 10, BMECs were rinsed once with DPBS (ThermoFisher) and lysed for 15 minutes at 4 °C using ice-cold radioimmunoprecipitation assay buffer (Rockland) supplemented with protease inhibitor cocktail (Pierce) according to the manufacturer's recommendation. Lysate, including cell membrane debris, was collected via scraping and centrifuged at max speed for 10 minutes at 4 °C. Supernatant was collected and the protein concentration quantified by bicinchoinic acid assay according to the manufacturer protocols (Pierce). Lysate was prepared for SDS-PAGE by mixing with 4X NuPAGE LDS Sample Buffer and NuPAGE Sample Reducing Agent. Samples were boiled at 70 °C for 10 minutes and loaded into 4 – 12 % Bis-Tris SDS-PAGE gels (ThermoFisher) at 10 – 20 µg protein / well. SDS-PAGE was run at 200 V using either NuPAGE MES SDS Running Buffer to resolve LRAT and CRBP1, or NuPAGE MOPS SDS Running Buffer to resolve STRA6. All running buffers were supplemented with NuPAGE Antioxidant according to manufacturer protocols. Samples were transferred at 30 V in NuPAGE Transfer Buffer to 0.45 micron polyvinylidene difluoride (PVDF) membranes (Amersham) for one hour. After transfer, PVDF membranes were washed three times at 5 minutes each in Tris-buffered saline with 0.1% Tween 20 (TBST). Membranes were blocked for 1 hour at room temperature in blocking buffer (5 % *w/v* non-fat dry milk dissolved in TBST) before overnight incubation at 4 °C with primary antibodies diluted in blocking buffer (**Table M2**).

After primary incubation, membranes were washed three times at 5 minutes each in TBST before incubation for 1 hour at room temperature in the dark with goat anti-rabbit IgG IRDye 680RD (Li-Cor, 1:5000 dilution, Cat # 926-68071) and donkey anti-mouse IgG IRDye 800CW (Li-Cor, 1:5000 dilution, Cat # 926-32212) secondary antibodies in blocking buffer. Membranes were washed three times for 5 minutes each in TBST and imaged using a Li-Cor Odyssey Imager.

Preparation of retinol, RBP and TTR

All-*trans* retinol (Sigma) and alpha-tocopherol (Sigma), which served as an antioxidant stabilizer, were dissolved in ethanol in equimolar concentrations and stored at – 80 °C. Concentrations of retinol (ROH) and alpha-tocopherol were determined by absorption using molar extinction coefficients of 52,480 $M^{-1}cm^{-1}$ at 325 nm and 3,260 $M^{-1}cm^{-1}$ at 292 nm, for ROH and alpha-tocopherol respectively. [15-3H(N)]-ROH (American Radiolabeled Chemicals), with an activity of 30 Ci / mmol, was supplied in an ethanol solution at 1 mCi / mL stabilized by alpha-tocopherol at a concentration of 1 mg / mL and stored at – 20 °C.

Recombinant human retinol binding protein IV (RBP) and mutant L63R/L64S retinol binding protein IV (muRBP) were produced in *E. coli* as inclusion bodies using the pTWIN system as described (58). Briefly, bacteria were lysed by sonication and the insoluble fraction collected by centrifugation. The inclusion body pellet was denatured and reduced in guanidine chloride supplemented with dithioerythritol (DTT), and then refolded in a cysteine / cystine oxidative buffer in the presence of excess all-*trans*-ROH. Refolded ROH-RBP or ROH-muRBP was purified on a chitin affinity column and released by intein-mediated self-cleavage. High quality ROH-RBP and ROH-muRBP preparations were confirmed by $A_{330/280}$ ratios of ~ 1.0 indicating ROH binding at 1:1 stoichiometry, and these preparations were stripped of retinol by successive liquid-liquid extractions via anhydrous diethyl ether. Stripped RBP was sparged with nitrogen to remove trace diethyl ether, and subsequently concentrated and buffer exchanged into phosphate buffered saline (PBS) via centrifugal filtration. Concentrations were quantified via absorption using an extinction coefficient of 40,400 $M^{-1}cm^{-1}$ at 280 nm. ROH-RBP and ROH-muRBP used for experiments were prepared as equimolar solutions 24 – 48 hours prior to use. Critically, confirmation of ROH binding was determined by absorption at 330 nm. The equilibrium dissociation constants for ROH to RBP and ROH to muRBP were both measured using fluorescence spectroscopy as described (58). ^{3}H-ROH-RBP and ^{3}H-ROH-muRBP were prepared identically using [15-3H(N)]-retinol.

Recombinant human transthyretin (TTR) and mutant I84A transthyretin (muTTR) were prepared using the pTWIN system as described (59). Briefly, protein was recovered from inclusion bodies by sonication in 8 M urea buffer, centrifugation to remove any insoluble material and rapid dilution of supernatant into Tris buffer to a final urea concentration of 4 M. The protein solution was applied to a chitin affinity column and allowed to refold on column. TTR was eluted by intein-mediated self-cleavage, and purified protein was dialyzed into PBS and concentrated via centrifugal filtration prior to storage at 4 °C. Concentrations were determined via absorption using an extinction coefficient of 77,600 $M^{-1}cm^{-1}$ at 280 nm. ROH-RBP-TTR and ROH-RBP-muTTR samples used for experiments were prepared 24 hours prior to use by mixing TTR or muTTR with stock solutions of ROH-RBP. Confirmation of TTR binding to ROH-RBP was determined by fluorescence anisotropy, where binding of TTR increases ROH anisotropy (Ex / Em : 330 / 460). ^{3}H-ROH-RBP-TTR and ^{3}H-ROH-RBP-muTTR complex were prepared identically using ^{3}H-ROH-RBP stock solutions.

Retinol cellular accumulation

BMEC monolayers cultured in 96-well plates were prepared as described and were used on Day 10 for retinol accumulation assays, performed at 37 °C on a shaker. BMECs from the same differentiation were seeded in parallel on Transwells to confirm quality via TEER measurements. Briefly, on Day 10 of the differentiation, cell media was aspirated from all wells and replaced with solutions composed of HBSS (ThermoFisher) and ROH, ROH-RBP, ROH-muRBP, ROH-RBP-TTR or ROH-RBP-muTTR. Unlabeled ROH was mixed one hour prior to use with a ^{3}H tracer prepared identically at a target final concentration of 5% ^{3}H-ROH. The actual concentration was quantified by liquid scintillation and tracer percentage adjusted accordingly. For each experiment, one 96-well plate was subdivided into 25 triplicate groups and loaded with the various ROH preparations. Wells were aspirated serially in triplicate every 5 minutes over the course of 2 hours, and each well was considered a biologically distinct replicate. Immediately after aspiration, each well was washed twice with HBSS and allowed to dry. All wells were lysed simultaneously by addition of 100 μL of ice-cold radioimmunoprecipitation assay (RIPA) buffer per well and incubated at 4 °C for 10 minutes.

Following lysis, each 100 μL lysate / RIPA mixture was placed into an individual liquid scintillation counting vial and measured for ^{3}H DPM. Briefly, vials were diluted with 10 mL of UltimaGold (PerkinElmer), shaken vigorously, and counted immediately three times for 5 minutes each on a Tri-Carb 2900TR Liquid Scintillation Counter (PerkinElmer). The average DPM from the three technical replicate readings was utilized as the readout. Tritiated samples were counted using a preset region of 0 – 18.6 keV, while carbon-14 samples, where required, were counted simultaneously using a preset region of 0 – 156 keV. Self-normalization and calibration (SNC) was performed using external standards prior to each data run. DPM were converted to cellular concentrations by using the specific activity of the tritiated retinol and an assumed cellular volume calculated by multiplying the average area of a cell by its average height.

Two-step RT-qPCR

BMECs were singularized on Day 8 and seeded on 24-well plates coated with 4:1:5 solution. On Day 10, BMECs were treated in parallel with the relevant ROH preparations for two hours. Each condition tested contained three biological replicates. Wells were aspirated after two hours and cells immediately incubated in Accutase for 30 minutes at 37 °C to promote dissociation. After 30 minutes, cells were collected in microcentrifuge tubes and centrifuged at 200 x *g* for 5 minutes. The supernatant was aspirated and samples were stored at – 80 °C. RNA from each individual sample was harvested independently using the Qiagen RNeasy Mini Kit (Qiagen) according to the manufacturer's instructions. To prepare cell lysate, samples were suspended in RNeasy kit Lysis Buffer and β-mercaptoethanol (1:100 *v/v*) and homogenized using QIAshredder columns (Qiagen). Samples loaded on spin columns were treated with DNase (Qiagen) to digest genomic DNA prior to RNA purification. RNA was eluted in molecular biology grade water (Corning) and quantified by UV-vis absorbance using an Eppendorf BioSpectrometer. Sample quality was confirmed by a clean absorbance scan with an $A_{260/280}$ ratio of > 1.8. 200 ng of purified RNA was immediately reverse transcribed at 37 °C for one hour in an S1000 Thermal Cycler (BioRad) via the OmniScript RT Kit (Qiagen) according to the manufacturer's instructions using RNase Out (ThermoFisher) and Oligo(dT)$_{12\text{-}18}$ primers (ThermoFisher). cDNA was used immediately or stored at 4 °C for no more than 24 hours prior to qPCR.

qPCR samples were prepared using 2X PowerUP SYBR Green (ThermoFisher), 10 ng of cDNA template prepared previously and PCR primers diluted in molecular biology grade water as described in **Table M3**.

Samples were loaded onto qPCR skirted plates (Agilent) and placed into an AriaMx Real-time PCR System (Agilent). Each run was initiated with a hot start at 95 °C for 15 minutes followed by 50 cycles consisting of a 15s denaturation step at 95 °C followed by a 60s amplification step at 60 °C. Fluorescence was analyzed after each cycle. After 50 cycles, the PCR product was melted and melt curves were inspected after each run to ensure only one peak was observed, corresponding to the appropriate PCR product as determined by agarose gel electrophoresis. Each condition analyzed contained 4 biological replicates with three technical replicates for each gene. An arithmetic mean Cq value was calculated from the technical replicates for each gene, and this value for BMECs treated with HBSS alone was used as the reference to calculate ΔCq values for each biological replicate of each gene for each ROH delivery condition. ΔCq values were converted to normalized expression quantities using an assumed 100% efficiency and ACTB as a housekeeping gene. Statistical analyses were performed on $\log_2$-transformed relative fold change quantities using one-way ANOVA followed by Dunnett's test with an assumed confidence interval of 95%. Log_2-transformed relative fold change data are presented graphically in box-and-whisker format.

Retinol BBB permeability assays

BMEC monolayers cultured on collagen / fibronectin 12-well Transwells were prepared as described. On Day 10 of the differentiation, TEER was optimal and BMECs were used for permeability assays. 550 μL of ROH-containing solution was applied in triplicate to the apical chambers and 1500 μL of HBSS was added into each basolateral chamber. ROH preparations were spiked with ^{14}C-sucrose (PerkinElmer) as a paracellular transport control. Samples collected from the apical and basolateral chambers over the course of the experiment totaled 10% of the bulk volume, and liquid removed from basolateral chambers was immediately replenished with the same volume of fresh HBSS. At t = 0, samples were collected from both apical and basolateral chambers. Plates were then placed at 37 °C on a shaker. Samples were collected from the basolateral chambers every 15 minutes. At t = 60 minutes, an additional sample was also collected from the apical chamber. Once removed, each sample was placed into a scintillation vial for counting. TEER was measured after the final sample collection at 60 minutes to verify integrity of the monolayer. At the conclusion of the assay, solutions in both the apical and basolateral compartments were aspirated and 100 μL of ice-cold RIPA buffer was added directly to the BMEC

Gene	Expected Product Size (bp)	Oligo Sequences 5' – Forward Primer – 3' 5' – Reverse Primer – 3'	Final Reaction Concentration (nM)
LRAT	149	AGCCTGCTGTGGAACAACTG GCCAATCCCAAGACTGCTGA	100
CRBP1	175	AGATCGTGCAGGACGGTGA CCCTTCTGCACACACTGGAG	100
STRA6	216	ACACACAGGACCAACCTTCGAG GAGCACAGGCATGAGCACCA	200
ACTB	218	CATCCGCAAAGACCTGTACG CCTGCTTGCTGATCCACATC	100

Table M3

qPCR primer information

monolayer to lyse the cells. After incubation at 4 °C for 10 minutes, cell lysate was scraped and collected for scintillation counting. Concentrations of tritium in the apical and basolateral chambers were calculated using the specific activity and well volume. Concentrations of tritium associated with the BMEC monolayers were calculated using the specific activity and the estimated volume of a single BMEC cell.

Data availability

Data will be shared by request to the corresponding author at regina.murphy@wisc.edu.

Supporting information

This article contains supporting information.

Acknowledgements

IMR90-4 iPSCs were a generous gift from Professors Eric Shusta and Sean Palecek. Images used to calculate the area of a single BMEC were collected and provided by Benjamin Gastfriend. Schematics of ROH transport at the BBB were created with BioRender.com. We thank Professor Eric Shusta and Ben Gastfriend for helpful discussions on BMEC differentiation and development of permeability assays.

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Conflict of interest

The authors declare that they have no conflicts of interest with the contents of this article.

Supporting information

$$C_V = V_D \frac{S_{A,t}}{S_{D,t\,=\,60\ min}} \qquad \textbf{Eq (S1)}$$

$$Pe_{app} = \frac{1}{A_F} m^{C_V} \qquad \textbf{Eq (S2)}$$

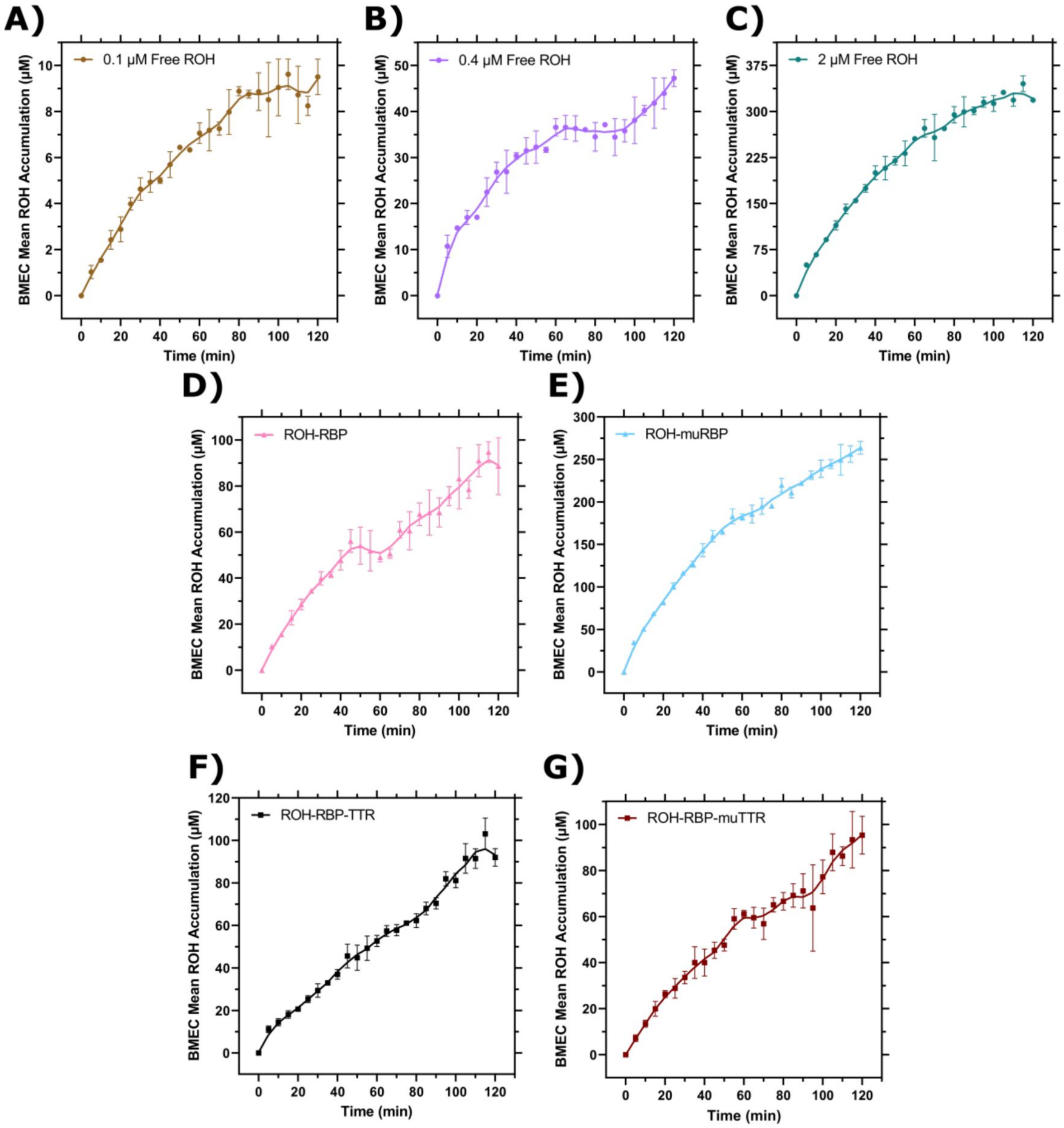

Figure S1

Individual BMEC ROH accumulation curves.

Measured DPM values were converted to cellular concentrations using the specific activity of ^{3}H-ROH, the ^{3}H-ROH: unlabeled ROH ratio (1:20) and the calculated cell volume. Error bars represent the standard deviation of three biological replicates. Data are fit by a smoothed 10 point LOWESS function in order to better view the kinetic regimes. **A)** 0.1μM Free ROH; **B)** 0.4μM Free ROH; **C)** 2 μM Free ROH; **D)** ROH-RBP; **E)** ROH-muRBP; **F)** ROH-RBP-TTR; **G)** ROH-RBP-muTTR.

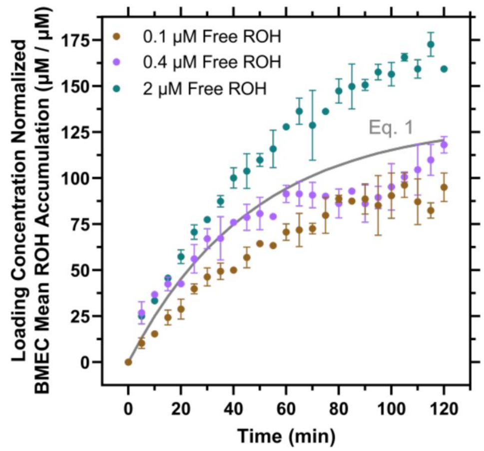

Figure S2

Mean ROH accumulation partitioning model.

Data from **Figure 4A** are normalized by the ROH concentration in the medium and fit by a simple partitioning model (**Eq. 1** in gray). If this model were correct, the data at all free ROH concentrations would collapse onto a single curve described by **Eq. 1**.

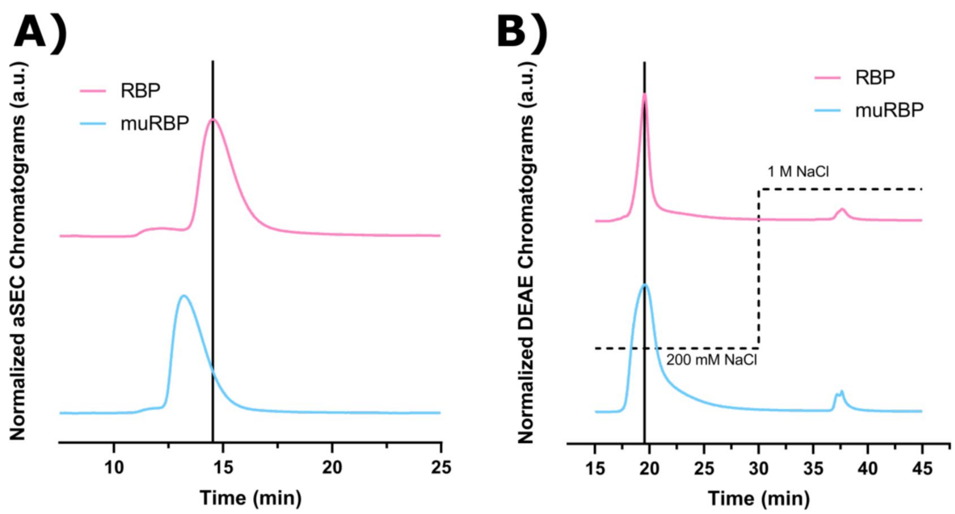

Figure S3

Normalized chromatographic analysis of wild-type RBP and L63R/L64S mutant RBP.

Spectra of both wild-type RBP (RBP), adapted from (58), and L63R/L64S mutant RBP (muRBP) were normalized to their respective max absorbance signals at 280 nm and overlaid on the same plot. The vertical bars represent the retention time corresponding to the max signal for RBP. **A)** ~ 15 μg of either RBP or muRBP in PBS (pH 7.4) was injected onto a TOSOH TSKgel BioAssist G2SWxl analytical size exclusion column operating at a flow rate of 1.0 mL/min of PBS (pH 7.4). **B)** RBP and muRBP were concentrated and buffer exchanged into AEX Buffer A (25 mM Tris, 1 mM EDTA, pH 8.0). The protein samples were filtered through a 0.22 μm filter (Millipore) and slowly applied by syringe to a GE HiScreen diethylaminoethyl (DEAE) column pre-equilibrated with AEX Buffer A. The sample was allowed to adsorb for 10 minutes, then re-equilibrated for 10 minutes with AEX Buffer A at a flow rate of 1.0 mL/min. A step salt gradient (dotted line) was applied by mixing AEX Buffer A with high salt AEX Buffer B (25 mM Tris, 1 M NaCl, 1 mM EDTA, pH 8.0) at 1.0 mL/min.

Sample	Replicate	Apical Counts x 10^{-3} (t0)	Apical Counts x 10^{-3} (t60)	Basolateral Counts x 10^{-3} (t60)	Lysate Counts x 10^{-3} (t60)	Missing Counts x 10^{-3} (t60)	% Missing
0.1 μM Free ROH	1	227.22	218.21	4.05	7.33	(2.38)	-1.0%
	2	235.00	218.08	4.77	7.26	4.89	2.1%
	3	238.90	218.61	4.68	8.23	7.37	3.1%
	4	233.22	219.94	4.98	8.83	(0.52)	-0.2%
0.4 μM Free ROH	1	326.57	295.66	7.81	13.70	9.40	2.9%
	2	326.87	291.44	8.98	13.43	13.02	4.0%
	3	327.87	292.72	9.28	13.44	12.43	3.8%
	4	333.21	300.92	10.26	15.28	6.75	2.0%
2 μM Free ROH	1	1,552.98	1,171.46	58.02	164.43	159.07	10.2%
	2	1,537.16	1,161.60	57.20	171.93	146.43	9.5%
	3	1,545.32	1,206.34	63.14	169.63	106.21	6.9%
	4	1,547.17	1,173.19	65.65	184.56	123.77	8.0%
ROH-RBP	1	1,799.84	1,675.54	60.68	49.51	14.11	0.8%
	2	1,840.65	1,688.63	62.44	53.72	35.85	1.9%
	3	1,848.14	1,681.03	62.71	54.53	49.87	2.7%
	4	1,819.80	1,680.83	66.05	63.90	9.01	0.5%
ROH-muRBP	1	1,534.63	1,427.66	59.51	67.42	(19.96)	-1.3%
	2	1,594.58	1,436.75	59.61	70.31	27.91	1.8%
	3	1,596.38	1,457.58	62.00	83.02	(6.22)	-0.4%
	4	1,555.79	1,441.91	62.05	74.77	(22.95)	-1.5%
ROH-RBP-TTR[1]	1	843.25	755.65	38.36	23.94	25.29	3.0%
	2	842.47	798.55	36.95	27.37	(20.40)	-2.4%
	3	826.47	778.79	36.47	23.65	(12.45)	-1.5%
	4	842.34	795.80	36.42	23.15	(13.03)	-1.5%
ROH-RBP-muTTR[1]	1	717.36	681.12	35.87	16.56	(16.19)	-2.3%
	2	724.06	679.06	36.40	15.12	(6.52)	-0.9%
	3	736.88	683.53	37.13	15.02	1.20	0.2%
	4	731.62	679.60	37.71	15.34	(1.03)	-0.1%

[1] TTR and muTTR samples were prepared with a target ^{3}H-ROH : unlabeled-ROH ratio of 1:40 in order to reduce the ethanol concentration required for 2X ^{3}H-ROH-RBP stocks. Precipitation of RBP was observed in 2X stocks prepared for use at 1:20 ratios. This problem only affected TTR and muTTR preparations because ROH-RBP and ROH-muRBP stocks could be prepared at 1X.

Table S1

Mass balances for ^{3}H-ROH

Sample	Replicate	Apical Counts x 10^{-3} (t0)	Apical Counts x 10^{-3} (t60)	Basolateral Counts x 10^{-3} (t60)	Lysate Counts x 10^{-3} (t60)	Missing Counts x 10^{-3} (t60)	% Missing
0.1 µM Free ROH	1	590.78	603.22	2.48	6.70	(21.62)	-3.7%
	2	608.26	602.21	2.28	5.86	(2.09)	-0.3%
	3	612.02	600.11	2.75	8.73	0.44	0.1%
	4	604.47	599.49	2.75	11.88	(9.65)	-1.6%
0.4 µM Free ROH	1	659.44	657.39	2.77	5.84	(6.56)	-1.0%
	2	665.84	647.03	4.25	4.32	10.24	1.5%
	3	666.92	645.73	3.21	3.94	14.04	2.1%
	4	666.63	663.05	3.63	5.43	(5.47)	-0.8%
2 µM Free ROH	1	649.77	631.68	3.35	4.39	10.35	1.6%
	2	645.77	633.23	3.33	5.15	4.05	0.6%
	3	646.95	634.52	3.15	4.55	4.73	0.7%
	4	648.13	628.08	3.23	5.64	11.19	1.7%
ROH-RBP	1	663.97	678.58	4.36	5.13	(24.11)	-3.6%
	2	663.64	668.09	5.16	6.18	(15.79)	-2.4%
	3	670.71	667.15	4.55	5.38	(6.37)	-0.9%
	4	662.58	680.76	4.97	9.01	(32.17)	-4.9%
ROH-muRBP	1	639.32	664.00	5.06	5.32	(35.06)	-5.5%
	2	651.86	660.14	4.83	5.11	(18.21)	-2.8%
	3	654.17	662.85	5.01	5.14	(18.83)	-2.9%
	4	642.95	662.28	5.57	5.70	(30.60)	-4.8%
ROH-RBP-TTR	1	656.36	656.31	3.38	8.04	(11.37)	-1.7%
	2	650.94	668.70	3.42	8.64	(29.82)	-4.6%
	3	657.11	666.45	3.35	7.47	(20.16)	-3.1%
	4	650.99	670.19	3.63	6.77	(29.59)	-4.5%
ROH-RBP-muTTR	1	602.46	619.32	2.77	5.59	(25.21)	-4.2%
	2	598.84	620.44	3.22	4.95	(29.77)	-5.0%
	3	611.35	616.41	3.03	5.52	(13.61)	-2.2%
	4	606.02	614.02	2.68	5.15	(15.83)	-2.6%

Table S2

Mass balances for ^{14}C-sucrose

Sample	ROH Pe_{app} (10^{-6} cm/s)	Sucrose Pe_{app} (10^{-6} cm/s)
0.1 μM Free ROH	3.75 ± 0.39	0.43 ± 0.03
0.4 μM Free ROH	4.65 ± 0.43	0.48 ± 0.04
2 μM Free ROH	7.68 ± 0.53	0.58 ± 0.03
ROH-RBP	4.76 ± 0.19	0.69 ± 0.04
ROH-muRBP	5.44 ± 0.17	0.76 ± 0.06
ROH-RBP-TTR	6.56 ± 0.30	0.58 ± 0.03
ROH-RBP-muTTR	7.61 ± 0.13	0.55 ± 0.04

[2] Pe_{app} is the mean apparent permeability (*N = 4 replicates*) of the BMEC monolayer and Transwell filter combined calculated by **Eq (S2)**, where A_F is the Transwell filter area and m^{C_V} is the fitted linear slope of the clearance volume (C_V) as a function of time. For samples displaying a lag-phase, slope was calculated only from the linear segment. C_V values were calculated by **Eq (S1)**, where V_D is the volume of the donor chamber and $S_{A,t}$ and $S_{D,t\,=\,60\,min}$ are the CPM signal in the acceptor chamber at time t and the CPM signal in the donor chamber at time $t = 60\ min$, respectively.

Table S3

Mean apparent permeability of ^{3}H-ROH and ^{14}C-sucrose[2]

Sample	Maximum percentage accumulated
0.1 µM Free ROH	16%
0.4 µM Free ROH	19%
2 µM Free ROH	28%
ROH-RBP	8%
ROH-muRBP	21%
ROH-RBP-TTR	9%
ROH-RBP-muTTR	8%

Table S4

Maximum percentage of fluid phase ROH accumulated by cells

Parameter	Eq 1. Fit	Eq 2. Fit
K_p (µM cell/ µM fluid)	131	99
k_1 (min^{-1})	0.021	0.027
K_p^* (µM cell/ µM fluid)	N/A	90
k_1^* (min^{-1})	N/A	0.014
t_{lag} (min)	N/A	0.1 µM Free ROH: N/A 0.4 µM Free ROH: 101 2 µM Free ROH: 7.8
c_{cell}^* (µM)	N/A	36
χ^2	17,930	2,470
Residual Sum of Squares (RSS)	41,000	3,700
Aikake Information Criteria (AIC)	17,934	2,480

Table S5

Partitioning model fits for BMEC free ROH accumulation

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Reviewer #1 (Public Review):

This study provides a novel in vitro model for the study of retinol transport across the human BBB by pairing iPSC-derived BMECs with the use of recombinant vitamin A serum transport proteins, RBP and TTR. Key findings of the paper include 1) the observation that the delivery mode of retinol affects its intracellular accumulation at the BBB but not its permeation across the BBB, 2) further highlighting that intracellular concentrations of retinol are also ensured by its efflux via its receptor STRA6 and 3) a potential novel role for TTR in retinol transport by upregulating LRAT mRNA expression, independently of RBP. Notably, the model appears to be more accurate than ones previously used (primary porcine BMECs) to study retinol delivery at the BBB, and could be used to study the retinol dysregulation at the BBB in neurodegenerative diseases (e.g. by using iPSC lines from NDD patients), something that miss in the paper.

Indeed, the major disappointment of this work is the clinical relevance that was highlighted in the Introduction but was never really studied in the end. iPSC from patients could be added to the study.

As a general comment, the study is well done however the introduction and the discussion as a bit long and do not get to the point of the work easily. Even sometimes losing the reader in many details (necessary here?). Less abbreviations would be appreciated for general readers.

Reviewer #2 (Public Review):

The manuscript by Est and Murphy tested the feasibility of using brain microvascular endothelial-like cells (BMECs) derived from induced pluripotent stem cells (iPSCs) as a model for studying retinoid uptake and transport across the blood-brain barrier (BBB). Establishing this experimental model is an important step towards obtaining greater mechanistic insight into the specificity of retinol trafficking between blood and retinoid-dependent tissues. The authors validated the iPSC-derived BMECs by detecting the expression of specific protein markers for BBB. They also demonstrated that BMECs form a tight barrier when cultured in a Transwell chamber, allowing for the quantification of permeability across the cells rather than through paracellular leakage. Finally, they confirmed the expression of the transporter (STRA6), binding protein (CRBP1), and enzyme (LRAT), which are key elements of the molecular machinery involved in the cellular uptake of circulating retinol. The carefully established model of the human BBB served as an experimental platform for the authors to investigate the uptake and transcellular transport of retinol. For this purpose, they compared the kinetics and efficiency of retinoid accumulation delivered to the cell as free retinol, retinol bound to serum retinol-binding protein (RBP), or retinol-RBP in complex with transthyretin (TTR), a physiological binding partner for retinol-loaded RBP.

Although the development and thorough characterization of the experimental model of the BBB have great value and meaningfully contribute to ongoing efforts to better understand the mechanisms of retinoid homeostasis, the premise and interpretation of cellular uptake appear controversial. In particular:

1. The authors assume that there is a significant fraction of free ROL, 20% for ROH/RBP and 7% for RBP/TTR complexes (summarized in Table 1). This implies that at the physiological concentration of ROH/RBP in the plasma of 2 uM, free ROL represents 0.4 uM. However, the concentration of free ROL is limited by its poor solubility in the aqueous phase, which is around 0.06 uM (Szuts EZ, 1991, Arch Biochem Biophys). Moreover, taking into account the large concentration of other potential nonspecific carriers for lipids, it is safe to assume that there is virtually no free ROH in the plasma. There is also an important physiological reason for the limited amount of free ROL. Its rapid and nonspecific partition into cells (also observed in this study) would work against the highly specific RBP/STRA6-dependent ROH uptake pathway, undermining its physiological function.

2. The advantage of the experimental system used in this report is that it allows for the assessment of the permeability across BMECs. Interestingly, the basolateral accumulation of ROH represented only a small fraction (1 - 1.5%) of the total ROH taken up by the cells. Moreover, the overall permeability was comparable regardless of the source of ROL added at

the apical side. However, a question remains: would the outcome of the experiment be different if the basolateral chamber contained an ROH acceptor (retinol-binding proteins) rather than Hank's balanced salt solution, to which the partition of ROL is limited by its water solubility? In fact, the maximum concentration of ROH on the basolateral side did not exceed 40 nM (Fig 5D and 7C), which is roughly the maximum water solubility of ROH. Thus, this experimental design limits extrapolation of the data to in vivo conditions.

3. The authors claim that transthyretin (TTR) increases BMECs permeability when compared to ROH/RBP. However, the mechanistic explanation for this phenomenon remains unclear. Do the authors imply the presence of a putative TTR receptor whose signaling could affect the efflux of ROL at the basolateral side of BMECs? TTR is an ubiquitous plasma protein. The concentration of TTR is tightly regulated and maintained between 300 - 330 mg/L. Therefore, it is questionable how TTR can serve as a signaling molecule modulating retinoid homeostasis in the brain.

4. Although overexpression of LRAT in response to increased uptake of ROH is well-documented, the postulate that TTR stimulates the expression of LRAT in an RBP-independent manner is puzzling, for the reasons mentioned in point 3. Moreover, LRAT is a highly efficient enzyme that operates under physiological conditions with substrate concentrations below the Km value. The rate of esterification is primarily limited by the intracellular transport of ROH to the ER. Therefore, without kinetic studies, it is unclear whether an increased number of LRAT copies (x2) would have a significant effect on the rate of accumulation of retinyl esters (REs).

5. The conclusion that cellular uptake of ROH is biphasic appears to be correct. However, the proposed interpretation of the mechanistic principles of this phenomenon is oversimplified. It assumes that loading CRBP1 with ROL to its capacity triggers the synthesis of REs. However, the saturation of CRBP1 with ROH is not required for REs formation. In fact, studies on CRBP1-deficient mice indicate that this protein is not necessary for the efficient esterification of ROL but rather affects the intracellular turnover of retinoids. It is likely that with increasing concentration of ROH, the specific and controlled mechanism of intracellular retinoid transport becomes saturated, allowing for spontaneous diffusion-driven partitioning of retinoids within cells.

Additional technical issues that could affect the experimental outcomes:

1. The formation of the ROH/RBP-TTR complex should be confirmed and purified using gel filtration to separate free TTR and ROH/RBP. Only fractions containing the complex should be used in the experiments. Assuming that the complex is formed with 100% efficiency is overly optimistic.

2. Reloading RBP with isotopically labeled ROH requires an additional purification step. Stripping ROL from the ROH/RBP complex with organic solvent (diethyl ether) is appropriate but relatively harsh, causing partial unfolding of a fraction of RBP. Therefore, assuming that 100% of stripped RBP remains functional and can be reloaded with ROH is inaccurate. Reloading apo-RBP with a stoichiometric amount of ROH without an additional purification step (e.g., ion exchanger) leads to an excess of free ROL and/or its nonspecific association with nonfunctional RBP fractions. Measuring absorbance at 330 nm is not sufficient proof of binding since free ROH also absorbs at the same wavelength.

Reviewer #3 (Public Review):

Vitamin A is critical for the development of the brain and for neuronal function and plasticity, however the mechanisms responsible for the uptake of retinol across the blood brain barrier (BBB) are currently not known. The authors investigate vitamin A uptake across the blood brain barrier using an in vitro model based on endothelial cells differentiated from human derived induced pluripotent stem cells. Using recombinant cargo proteins and radioactive tracers the authors then propose a mechanism and a kinetic model for the uptake of retinol across the BBB that requires serum retinol binding protein 4 (RBP4 or RBP) and its receptor stimulated by retinoic acid 6 (STRA6). The results support a concentration dependent mechanism of transport combining a rapid fluid-phase retinol and a slower directed RBP-complexed retinol across the BBB. The data also hint at the potential regulatory roles of TTR on this process independent of its interaction with RBP.

Strengths:
The studies are rigorous and careful and the authors consider free retinol uptake from the fluid-phase in addition to evaluating RBP-TTR and RBP-STRA6 interactions.
The antibody to STRA6 is validated.
The experiments performed are clearly described.

Weaknesses:
The results presented do not offer significant new information regarding the uptake of retinol by tissues beyond what is known and published using genetic, structural and biochemical approaches.
The use of the iPSC-derived BBB model is potentially interesting but this could have been complemented by a thorough genetic dissection of the cellular factors required for the uptake, transcellular transport, and secretion of retinol by the brain endothelial cells.
The conclusions derived are not well supported by the data presented.
It is difficult to infer a mechanism or to derive a meaningful conclusion regarding the in vivo relevance of the results presented.

Author Response

We thank the reviewers for their very thorough and detailed comments as well as the overall positive reception of the work. Additionally, the reviewers provided excellent detailed suggestions for future work.

> *Specific response to Reviewer 1:*
>
> *"Indeed, the major disappointment of this work is the clinical relevance that was highlighted in the Introduction but was never really studied in the end. iPSC from patients could be added to the study."*

We completely agree that it would be very exciting to use patient-derived iPSC in the platform that we describe in this manuscript. We recognize that extensive work to characterize and validate BMECS differentiated from patient-derived iPSCs would be required, including validating BBB-like properties, before retinol transport data could be collected and interpreted. This work is beyond the scope of the current manuscript. We hope that in the future the in vitro model we describe in this manuscript will be used for exactly this type of clinically relevant application.

> *Specific response to Reviewer 2:*
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> *"1) The authors assume that there is a significant fraction of free ROL, 20% for ROH/RBP and 7% for RBP/TTR complexes (summarized in Table 1). This implies that at the physiological concentration of ROH/RBP in the plasma of 2 uM, free ROL represents 0.4 uM. However, the concentration of free ROL is limited by its poor solubility in the aqueous phase, which is around 0.06 uM (Szuts EZ, 1991, Arch Biochem Biophys). Moreover, taking into account the large concentration of other potential nonspecific carriers for lipids, it is safe to assume that there is virtually no free ROH in the plasma. There is also*

> *an important physiological reason for the limited amount of free ROL. Its rapid and nonspecific partition into cells (also observed in this study) would work against the highly specific RBP/STRA6-dependent ROH uptake pathway, undermining its physiological function."*

The reviewer raises an important point that we considered carefully during the design of the research. As the reviewer says, Szuts (1991) reported retinol (ROH) solubility of ~0.06 μM (range of 0.03 – 0.11 μM). Szuts defined ROH solubility as 'the amount of dissolved solute in equilibrium with its solid state…includ[ing] all its dissolved forms (monomers, multimers, and micelles)'. We are using a definition of 'free' ROH as 'ROH not bound to protein'; in our work 'free' ROH could include retinol multimers and micelles, which likely do exist under our experimental conditions. (We did not see any evidence of solid ROH.) That said, we calculate that the concentration of free ROH (ROH not bound to protein) is ~0.14 μM when both RBP and TTR are present. In more complex biological mixtures containing other ROH carriers, the concentration of unbound ROH is expected to be lower, in agreement with the reviewer.

One key point is that the free ROH concentration depends on the experimental setup, and must be correctly accounted for. For example, in some of the literature investigating STRA6-mediated uptake and signaling in vitro, purified ROH-RBP is used as the retinol source and samples do not include TTR. In such a case, the unbound ROH concentration in an equilibrated sample is anticipated to be significantly higher than the physiological concentration. Our investigation demonstrates that unbound ROH can accumulate intracellularly; thus, failure to include TTR and/or to account for the action of unbound ROH could lead to errors in mechanistic interpretation of experimental studies on retinol transport into cells or across barriers such as the BBB.

> 1. *"However, a question remains: would the outcome of the experiment be different if the basolateral chamber contained an ROH acceptor (retinol-binding proteins) rather than Hank's balanced salt solution, to which the partition of ROL is limited by its water solubility?"*

We agree with the reviewer that it would be very interesting to determine whether retinol permeability changes in the presence of RBP and/or TTR on the basolateral side. This is a logical next step and can readily be performed in the Transwell setup. We chose not to do this for this project because we wanted to compare our setup with other in vitro models (e.g., with porcine BMECs) where no retinol-binding proteins were present basolaterally.

> 1. *"The authors claim that transthyretin (TTR) increases BMECs permeability when compared to ROH/RBP. However, the mechanistic explanation for this phenomenon remains unclear. Do the authors imply the presence of a putative TTR receptor whose signaling could affect the efflux of ROL at the basolateral side of BMECs? TTR is an ubiquitous plasma protein. The concentration of TTR is tightly regulated and maintained between 300 - 330 mg/L. Therefore, it is questionable how TTR can serve as a signaling molecule modulating retinoid homeostasis in the brain."*

We disagree with the reviewer about the TTR concentration. Per Johnson et al (Clin Chem Lab Med 2007, 45:419-426), TTR concentration varies with age, gender, inflammation and nutritional status, with typical concentrations for adults ranging from 150-450 mg/L. We were surprised at our observations that TTR enhanced ROH permeability across BMECs and that LRAT expression increased in the presence of TTR. We do not currently have a mechanistic interpretation and agree with the reviewer that further exploration of these tantalizing observations is warranted.

> *"Additional technical issues that could affect the experimental outcomes: The formation of the ROH/RBP-TTR complex should be confirmed and purified using gel filtration to separate free TTR and ROH/RBP. Only fractions containing the complex should be used in the experiments. Assuming that the complex is formed with 100% efficiency is overly optimistic."*

We respectfully disagree with the reviewer regarding using gel filtration to isolate TTR/ROH/RBP complexes. Any such isolated complexes will fairly rapidly re-equilibrate so that some protein and some ROH is unbound. It is important to note that we do not assume that the complex is formed with 100% efficiency. In fact, on the contrary, we explicitly take into account the distribution of materials (free TTR, free RBP, free ROH, RBP-ROH, TTR-RBP-ROH) in any sample; values are reported in the manuscript. This issue is also relevant to the first point raised by the reviewer. We routinely validated binding of ROH to RBP by FRET and ROH-RBP to TTR by fluorescence anisotropy.

> *"Reloading RBP with isotopically labeled ROH requires an additional purification step. Stripping ROL from the ROH/RBP complex with organic solvent (diethyl ether) is appropriate but relatively harsh, causing partial unfolding of a fraction of RBP. Therefore, assuming that 100% of stripped RBP remains functional and can be reloaded with ROH is inaccurate. Reloading apo-RBP with a stoichiometric amount of ROH without an additional purification step (e.g., ion exchanger) leads to an excess of free ROL and/or its nonspecific association with nonfunctional RBP fractions. Measuring absorbance at 330 nm is not sufficient proof of binding since free ROH also absorbs at the same wavelength."*

We produced RBP by refolding of guanidine-denatured RBP in an excess of ROH to ensure near 100% ROH loading. High quality refolded RBP can qualitatively be determined by examination of the A330/280 absorbance ratio, which should be ~1.0. We then extract ROH to completion by diethyl ether to produce pure apo-RBP (ROH-free). We utilized this diethyl-ether stripped apo-RBP stock for all future characterizations, including binding to ROH and TTR. We found our stripped apo-RBP was a suitable replacement for serum sources in every biophysical assay performed. Reloaded ROH-RBP elutes as a single peak on ion exchange chromatography, indicating the vast majority of stripped RBP is available for ROH binding. We provide detailed information about RBP characterization in Est and Murphy, Prot. Exp. Purif. (2020), to which the interested reader is referred.